

Passenger-oriented rolling stock scheduling in the metro system with multiple depots: Network flow based approaches

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ABSTRACT

This study investigates a rolling stock scheduling problem on a metro line with multiple depots. Two novel optimization models, i.e., an arc-based and a path-based network-flow models, are formulated with the aim of improving the service level and reducing the operation cost simultaneously, in which the flexible train composition mode is also taken into consideration to well match the transport capacity and time-varying passenger demand. To solve the proposed models, a branch-and-price (B&P) approach is designed to find the near optimal operation schemes, in which the column generation is used to solve the relaxed problem at each node of the searching tree, where a dynamic programming approach is embedded to solve the pricing sub-problem associated with each depot to generate promising paths (columns) for each rolling stock unit, and then the branch-and-bound (B&B) procedure is incorporated to find integral solutions. To test the performance of the proposed approaches, a series of numerical experiments are conducted both on small-scale and real-life cases of the Beijing metro Batong line with historically recorded passenger data. The computation results have verified the improved operational efficiency and a better service level of the solutions found by our proposed approaches.

1. Introduction

The metro system has advantages over other transportation modes in the inter-city transportation in terms of large capacity, punctuality and comfortable traveling experiences. It has been being developed rapidly in some mega-cities (e.g., Beijing, Shanghai) of China. Up to 2022, the total mileage of the metro network in Beijing has reached 783 km, which already becomes the backbone of city commuting. To well operate the metro system, transportation planning plays a significant role in providing the efficient operation schemes. In general, the transportation planning process consists of multiple components on various decision levels (i.e., strategic, tactical, and operation levels), such as line planning, train scheduling, rolling stock circulation planning, crew scheduling and train rescheduling (Lusby et al., 2011; Wang et al., 2018), which are often solved in a sequential and phase-wise manner, illustrated in Fig. 1.

As displayed, train scheduling is a core component in the process of operations, which specifies the train arrival and departure time at each station according to the input line planning scheme. With the given schedule as input, the rolling stock circulation planning then assigns the available rolling stock units to undertake all scheduled services with the goal of reducing the total operation cost. Two operation modes can be taken into account in the process of rolling stock circulation planning, i.e., the fixed train composition mode and the flexible train composition mode. The former is a much simple and commonly used operation mode

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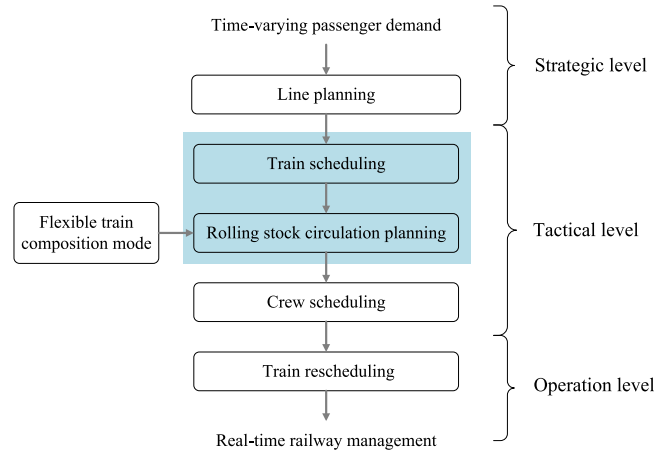


Fig. 1. Transportation planning process in the metro system.

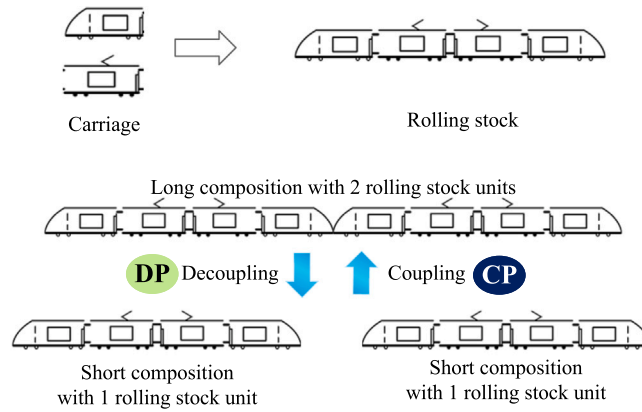


Fig. 2. Illustration of terms for flexible train composition mode.

used for most of metro lines at present, in which no change of composition is permitted. When the passenger demand distribution is uneven, this mode, however, might lead to a waste of transport capacity on some segments with less passengers. Thus, a more flexible operation mode, i.e., the flexible train composition, is needed in order to better match the passenger demand distribution and transport capacity, which can change the train composition at some designated stations by coupling and decoupling activities. We next introduce some terms for descriptive convenience. As shown in Fig. 2, a rolling stock unit refers to a least number of carriages with the self-propelled engine, which corresponds to the smallest train composition. In the operations, two or more rolling stock units can be coupled to form a large composition with more capacity, and a large composition can also be decoupled into some small train compositions if necessary. With the development of communication and control technologies, the coupling and decoupling activities can be completed in several minutes, which almost does not affect the normal operations, and moreover it is conducive to planning the operation scheme that can better match the passenger distribution and the capacity supply with less operation costs. In Fig. 2, the symbols “CP” and “DP” denotes the coupling and decoupling activities.

We note that the flexible train composition mode will necessarily cause the complexity in the operation planning process, in which different components need to be taken into consideration collaboratively (e.g., the rolling stock schedule, coupling/decoupling activities, passenger demand distribution, etc.), focusing on achieving the system optimization. The passenger-oriented rolling stock scheduling with flexible compositions and multiple depots (PRSFCD) can be formulated as a network-flow-based model in a space–time network, where rolling stock units between different depots are regarded as commodities. With the space–time network representation, different operations, such as coupling/decoupling activities, turnaround operations, pull-in (to depots) and pull-out (from depots) operations, can be denoted by space–time arcs in such a network. The objective of the problem is then to generate a set of space–time paths of rolling stock units that satisfy all operation constraints and passenger demand with the minimum operation cost and passenger waiting cost. The main contributions of this study can be summarized as follows.

(1) We propose a mixed-integer programming (MIP) model for PRSFCD, which can be characterized by the space–time trajectory optimization process in an extended space–time network. The objective aims to minimize the fixed cost associated with rolling stock units and waiting passengers, by imposing the trip covering constraints, flow conservation constraints, and passenger

flow assignment constraints. Compared with the existing methods, this study introduces the trip covering constraints to represent the rolling stock assignment modes, and integrates the time-varying passenger flow into the optimization model by introducing flow balance constraints. The safety headway constraints are also introduced to guarantee the requirements for operation safety.

(2) The rolling stock path-based reformulation and a branch-and-price algorithm are developed to solve the problem of our interest. In detail, the path-based model is firstly relaxed into a linear programming (LP), which is then transformed into a restricted master problem (RMP) and a series of rolling stock path finding sub-problems. The column generation algorithm is designed to solve the LP relaxation model at each node of the searching tree, which is embedded in the branch-and-bound algorithm for finding the integral solution to our model. At the same time, a dynamic programming (DP) algorithm is incorporated to find promising space–time paths (i.e., columns) for different rolling stock units in parallel.

(3) To validate the proposed models and algorithms, a series of numerical experiments are conducted for a small-scale instance and a series of real-world instances on the Beijing metro Batong line. The experimental results show that the proposed approach can find high-quality solutions for the considered problem, and the algorithm can well balance computational complexity and solution accuracy of the considered problem. Moreover, the performance analyses with respect to critical parameters (e.g., different numbers of depots, rolling stock capacities, and passenger demand scenarios) demonstrate the robustness of the proposed method.

The remainder of this paper is organized as follow: Section 2 gives a brief literature review in the field of train operation planning. Section 3 introduces a detailed problem statement and the representation of an extended space–time network. Two network-flow optimization models, i.e., an arc-based model and a path-based model, are formulated in Section 4 for the problem in consideration. Then, a branch-and-price algorithm is designed in Section 5 to solve the problem of interest. Section 6 conducts a variety of numerical experiments to verify the effectiveness and performance of the proposed methods. Finally, some conclusions and further research directions are given in Section 7.

2. Literature review

Practically, train scheduling and rolling stock circulation planning are two critical procedures in planning the operation schemes for the metro system, in which the time-varying passenger demand is usually incorporated in order to improve the service level (Ahuja et al., 2005; Harrod, 2012; Niu and Zhou, 2013; Sun et al., 2014; Barrena et al., 2014; Niu et al., 2015). We next aim to review the state of the art in this field from three aspects, including passenger-oriented train scheduling, joint optimization of train scheduling and rolling stock circulation planning problem, and some related studies of the flexible train composition mode.

The existing literature on the passenger-oriented train scheduling problem focuses on specifying the train departure and arrival time at each station with safety headway based on the time-varying passenger demand data, which has been a hot topic for metro operation planning in past decades. Compared with the traditional train scheduling problem, the passenger-oriented train scheduling highlights the effect of passenger distribution on the operation schemes (Schiewe and Schöbel, 2020; Martin-Idradi and Ropke, 2022), which aims to guarantee the higher service level in the metro system. In this problem, the train departure frequency and headway should be closely related to the time-varying passenger arrival volumes at each station, and the total passenger waiting time and train operation cost are in general treated as the objective functions (Barrena et al., 2014; Canca et al., 2018; Kidd et al., 2019). Two major solution approaches have been developed for the passenger-oriented train scheduling problem in the literature. The first approach refers to the job-shop-scheduling modeling framework with big- M constraints, which can well characterize the operation schemes explicitly (Liu and Kozan, 2011; Veelenturf et al., 2017; Szymula and Bešinović, 2020). It is difficult nevertheless to decompose this type of model into easily solvable sub-problems due to the complex coupling relationships. The second approach refers to the space–time-network based methods, which characterize the movements of trains and passengers as network flows in the space–time network (Yin et al., 2016; Yang et al., 2020; Zhang et al., 2022; Yao et al., 2023). Some classical decomposition methods, such as Lagrangian relaxation (LR), alternating direction method of multipliers (ADMM), column generation (CG), and Benders decomposition (BD), can be embedded into such modeling frameworks, and sub-problems can be solved by the shortest path algorithms without any assistance of commercial solvers (Shang et al., 2019; Zhan et al., 2021).

The joint optimization of train scheduling and rolling stock circulation planning aims to simultaneously find the operation scheme of the available rolling stock units in the rail system. In the early studies, most researchers usually focused on the joint optimization for the ordinary railway systems (Fioole et al., 2006; Peeters and Kroon, 2008). With the rapid development of the metro system, this topic has also been attracting much attention from researchers, in which two types of costs are mainly involved, i.e., the passenger cost and the operation cost. Other relevant costs, such as the movement cost of empty train and the penalty for passenger comfortable capacity, are also taken into consideration (see Cadarso and Marín, 2012). Considering the time-varying passenger demand (as a discrete piece-wise linear function), Wang et al. (2018) developed a multi-objective mixed-integer nonlinear programming (MINLP) in order to model the train scheduling and rolling stock circulation planning on a metro line. They also proposed two effective approaches, i.e., the iterative non-linear programming method and the mixed-integer linear programming (MILP) transforming method. Additionally, from the energy-saving perspective, Wang et al. (2021b) investigated a joint optimization of the energy-efficient timetabling and rolling stock circulation planning for metro lines based on automatic train operation levels, and moreover, a real-time train scheduling model under disruptions was proposed with the joint optimization of train rescheduling and rolling stock circulation planning on a metro line (Wang et al., 2021a). Based on the discrete space–time network, Yue et al. (2017) focused on a bi-level optimization method with consideration of time-dependent passenger demand, which optimizes the train timetable on the upper level and the rolling stock utilization plan on the lower level. A heuristic algorithm based on simulated annealing (SA) was used to solve the model and generate an approximate optimal solution. Mo et al. (2019) investigated an integrated model to simultaneously generate train schedule and rolling stock use plan with the representation of departure/arrival binary variables

Table 1
Comparison between closely related studies in the literature.

Studies	Model type	Model consideration					Solution method	Largest case study scale
		Flexible composition operation	Rolling stock circulation	Train schedule	Passenger demand	Multiple depots		
Abbink et al. (2004)	General MIP model	Y	Y	Fixed	N	N	CPLEX 7.0	Trip: N/A Rolling stock units: 188
Fioole et al. (2006)	General MIP model	Y	Y	Fixed	Trip demand	N	CPLEX 9.0	Trips: 665 Rolling stock units: 85
Cacchiani et al. (2010)	Arc-based model & Path-based model	Y	Y	Fixed	Trip demand	N	CG	Trips: 660 Rolling stock units: N/A
Cacchiani et al. (2013)	Arc-based ILP model	Y	Y	Fixed	Trip demand	N	Heuristic LR	Trips: 1190 Rolling stock units: 148
Lin and Kwan (2016)	Arc-based model & Path-based model	Y	Y	Fixed	Trip demand	N	B&P	Trips: $\approx$ 500
Pan et al. (2023)	Space-time network	Y	Y	Y	Dynamic	N	TS	Rolling stock units: 260
Zhou et al. (2022b)	General MILP model	Y	Y	Y	Dynamic	N	Diving	Trips: 99
This paper	Space-time network	Y	Y	Y	Dynamic	Y	CG	Rolling stock units: 55
							VNS	Trips: 186
								Rolling stock units: 41
								Trips: 260
								Rolling stock units: 20

Notes for Table 1:

(1) Rolling stock circulation: Joint optimization with the rolling stock circulation (Y).

(2) Train schedule: Fixed train schedule as input data (Fixed); Joint optimization with the train scheduling problem (Y).

(3) Passenger flow: Each trip is expected with a certain of passenger demand, which is often measured in number of seats (Trip demand); Dynamic passenger demand with the time-varying flow at each station (Dynamic); No consideration of passenger flow (N).

(4) Solution method: Column generation (CG); Lagrangian relaxation (LR); Branch-and-price (B&P); Topological sorting (TS); Dynamic programming (DP); Variable neighborhood search (VNS).

(5) Largest case study: Not available (N/A). That is, the exact numbers are not given directly as parameters.

at each station, in which train turn-around constraints, train circulation constraints, and passenger demand constraints were also considered.

In the real-world operations, the flexible composition mode is regarded as an efficient approach to improve the utilization of available rolling stock units, which can be embedded into operation planning to reduce the redundant capacity supply under the promise of the guaranteed service level. The problem of optimizing the flexible composition mode can be seen as allocating an appropriate number of rolling stock units to each trip. In the literature, the flexible composition mode is mainly applied to two typical scenarios, i.e., rolling stock maintenances and passenger-oriented operations. For the first case, [Maróti and Kroon \(2005\)](#) formulated a transition model based on the multiple commodity flow framework, and pointed out that the model with a single train unit is polynomially solvable, but the model with multiple train units is an NP-hard problem. Moreover, they also proposed an interchange model for train unit maintenance requirements towards a maintenance facility, and the computational complexity of the maintenance routing problem was analyzed in [Maróti and Kroon \(2007\)](#). [Lin and Kwan \(2016\)](#) proposed a branch-and-price method for the maintenance problem, by considering the additional real-life rolling stock requirements, such as train unit-type compatibility relations, the location banned for coupling/decoupling, and the combination of specific coupling upper bounds. [Zhong et al. \(2019\)](#) considered a problem with maintenance requirements under multiple types of train compositions and proposed a two-stage heuristic decomposition approach to solve the developed model. Based on a trip sequence graph, [Gao et al. \(2020\)](#) proposed two integer linear programming (ILP) models (path-based and arc-based models) to characterize the maintenance requirements of train units in a railway network and periodic characters of trip sequences, and then a branch-and-price-based algorithm was designed to solve the path-based model efficiently.

In the second scenario, train operations with the flexible composition mode can match the non-balanced spatial-temporal distribution of passenger flows much more efficiently. Along this line, [Abbink et al. \(2004\)](#) proposed an optimal allocation model to allocate train types and sub-types on rail lines based on the passenger volumes. Compared with the manual planning method, the result solved by CPLEX 7.0 had a significant service improvement within a shorter computing time. [Fioole et al. \(2006\)](#) proposed a rolling stock circulation model with the consideration of coupling and decoupling passenger trains with the given departure and arrival time at stations and the expected number of passengers. [Cacchiani et al. \(2010\)](#) developed an arc-based and a path-based ILP formulation to solve a real-world rolling stock assignment problem that can fulfill the seat requests of all passengers. The proposed model was solved by the column generation (CG) based method derived from the convex hull of the variant of the knapsack polytope. Later for a similar problem, [Cacchiani et al. \(2013\)](#) proposed a faster heuristic method based on the Lagrangian relaxation (LR) of a natural formulation, and the solution can be found by solving a sequence of assignment problems. In addition, [Cacchiani et al. \(2019\)](#) proposed a heuristic method based on a lower bound and proved that it was an NP-hard problem even if no constraints were imposed on the number of train units assigned to a trip. Based on the first-in-first-out rule for giving the same priority to a person arriving at the station simultaneously, [Shi and Li \(2021\)](#) proposed a MILP model to determine headway and vehicle capacity (as well as the number of vehicle units) to minimize the total transit cost. Benefited from the upper bound (UB) and lower bound (LB) with theoretical proprieties, the search state dimensions of a typical DP can be significantly reduced.

Finally, in order to highlight the innovations of this paper, Table 1 makes a detailed comparison regarding some closely related studies that consider the flexible train composition mode, in terms of model consideration, model type, solution algorithm and case study scale. The key differences between our research and existing ones can also be summarized as follows: (a) *Passenger demand*: compared with the input passenger volumes associated with the number of seats in fixed trips (i.e., trip demand) ([Fioole et al., 2006](#); [Cacchiani et al., 2010, 2013](#); [Lin and Kwan, 2016](#)), this paper formulates the rolling stock scheduling process based on

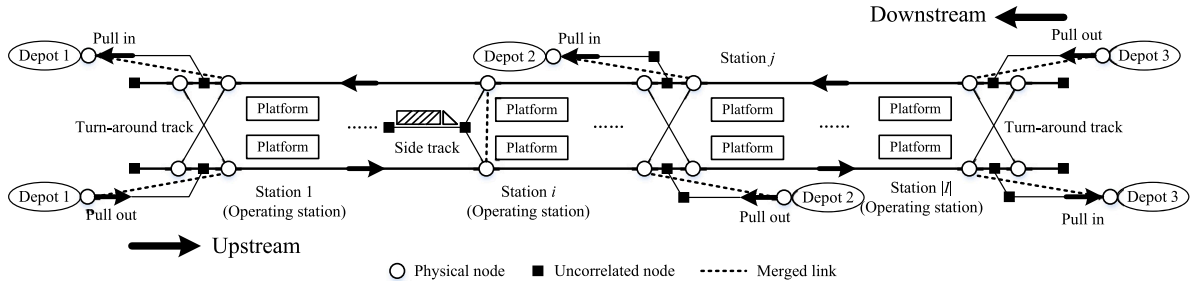


Fig. 3. Metro line with multiple depots.

the dynamic inbound and outbound passenger data associated with stations and timestamps, which can characterize the passenger flowing process more precisely and avoid difficulties in generating demand for unfixed train schedules. (b) *Case study scale*: Some previous studies (Fioole et al., 2006; Cacchiani et al., 2010, 2013; Lin and Kwan, 2016) can achieve 500–1000 trips in the largest case study, thanks to only finding connection schemes between fixed trips in a relative small problem scale. In comparison, when the train scheduling is integrated into the problem (Zhou et al., 2022b; Pan et al., 2023 and this paper), the problem scale will increase dramatically due to a large amount of space–time arc selections, resulting in less numbers of solvable trips (i.e., 100–200 trips).

3. Problem statements

This paper considers a bidirectional metro line with multiple depots, as shown in Fig. 3. Each depot is connected with the operating stations along the metro line. A certain number of rolling stock units are pre-stored in different depots, and the total number of in-service rolling stock units should not exceed the number of available vehicles in the system. In consideration of individual rolling stock units with diverse pull-in/out depots and time windows (Wang et al., 2022), one rolling stock unit is denoted by $k \in \mathcal{K}$, where $\mathcal{K}$ is the set of rolling stock units. At the end of daily operations, online rolling stock units are required to return to their home depots. For clarity, the total number of stations is denoted by $|I|$, and the turn-around operations occur on turn-around tracks located at both ends of the line or cross-tracks at the operating stations. In the problem of our interest, rolling stock units could be coupled or decoupled at some designated stations (i.e., operating stations) with turn-around tracks or side-tracks, while the coupling/decoupling operations are prohibited at other common stations.

In metro systems, the passenger distribution often shows a notable dynamic characteristic at different stations, i.e., the number of passengers in one direction is significantly larger than that in another direction during peak hours. Then, the passenger demand at each station also presents an “uni-modal” distribution during peak hours in both directions. On such a line, if the same number of rolling stock units are scheduled for both directions in the fixed train composition mode, the over-strained and redundant capacities, will possibly occur in the upstream and downstream directions. In comparison, when imposing the flexible composition mode, dispatchers could dynamically adjust the in-service fleet size in accordance to the passenger distribution, which simultaneously enables to guarantee the service level and reduce the redundant operation cost to a great extent.

This study focuses on investigating a passenger-oriented rolling stock scheduling problem with the flexible compositions on a metro line possessing multiple depots. The task is to determine the space–time trajectories of all in-service rolling stock units, in which the coupling and decoupling activities are incorporated to satisfy the time-varying passenger demand. The input data consist of the time-dependent passenger demand, the topological structure of the focused metro line, the number of available rolling stock units in each depot, etc. The goal is to minimize both the operation cost for each train unit as well as the total passenger waiting cost, and generate a high-quality rolling stock scheduling scheme with specific coupling/decoupling operations. To characterize the spatial and temporal movements of the involved trains and passengers in the metro system, the following discussion firstly introduces the space–time network representation method for the modeling process.

3.1. Space–time network for scheduling rolling stock

The space–time network is a visualization method to well capture the movements of rolling stock units in the metro system. To construct a space–time network, the considered time horizon is firstly discretized into a set of timestamps by a time granularity, denoted by $\mathcal{T} = \{0, 1, \dots, |\mathcal{T}|\}$. On the metro line, each physical node (e.g., station, depot) will then be extended to a series of space–time vertices over the discrete timestamps. A space–time vertex can be denoted by a unified form (i, t) , showing the spatial and temporal status of node i at timestamp t . The rolling stock movements can be represented by directed arcs in between space–time vertexes. Then, set $\mathcal{V}^R$ represents the set of space–time vertices (i.e., $(i, t) \in \mathcal{V}^R$). Particularly, some space–time vertexes from set $\mathcal{V}^R$, denoted as depot origin vertex $(d_{ori}(k), t_0)$ and depot destination vertex $(d_{des}(k), t_{|\mathcal{T}|})$, are used to show the source and sink states for a rolling stock unit k . Then, we use notation (i, j, t, t') to represent a space–time arc from the space–time vertexes (i, t) to (j, t') , and each space–time arc is associated with a pre-given cost. The involved arcs in the space–time network can be categorized into the following five types.

- **Traveling arc:** The traveling arcs indicate the movements of rolling stock units on the segments, e.g., a traveling arc (i, j, t, t') represents the movement from station i to station j in time interval $[t, t']$. The cost of a traveling arc often refers to the fixed transportation cost on the corresponding segment (e.g., energy consumption).

- **Waiting arc:** The waiting arcs indicate the dwell operations at physical nodes (i.e., stations or depots), e.g., a waiting arc $(i, i, t, t + 1)$ represents the dwell operation at physical node i in time interval $[t, t + 1]$. A station waiting arc denotes the dwell activity for loading/unloading passengers, and a depot waiting arc denotes the parking state of rolling stock units without any duties. Particularly, the rolling stock storage activity (i.e., rolling stock is not put into operation) is represented by a space–time path with a series of depot waiting arcs.

- **Pull-in/out arc:** The pull-in and pull-out arcs indicate the operations of entering and leaving a depot, respectively. Let $(d_{ori}(k), t)$ denote a space–time depot origin vertex of rolling stock unit k , and (i, t') denote a space–time vertex of its connected station i . If there is a pull-out arc $(d_{ori}(k), i, t, t')$, it indicates the operation of leaving depot $d_{ori}(k)$ at time t and arriving at station i at time t' . The pull-in arc can be defined similarly. In addition, a fixed cost is particularly given to this type of arcs to discourage frequent pull-in/out operations.

- **Turn-around arc:** The turn-around arcs denote the turn-around operations between a pair of stations connected with turn-around tracks. For instance, a turn-around arc (i, i, t, t') represents the turn-around activity at station i in time interval $[t, t']$. In general, the onboard passengers should get off trains before this type of operations. Particularly, the maximum turn-around time can be enlarged to construct more turn-around arcs for satisfying rolling stock flow conservation relationships for some peak-hour trips in upstream and downstream directions.

We denote the space–time network by $\mathcal{G}^R = (\mathcal{V}^R, \mathcal{A}^R)$, where $\mathcal{V}^R$ and $\mathcal{A}^R$ represent sets of space–time vertices and space–time arcs, respectively. In the space–time network $\mathcal{G}^R$, the rolling stock units in each depot will be scheduled to generate the optimal operation schemes. For modeling convenience, we next introduce the concepts regarding trips and paths that are frequently used in the formulations.

- **Traveling trip:** A traveling trip can be defined as a sequence of space–time vertices that are connected by traveling arcs and waiting arcs, from a departure vertex at one operating station to an arrival vertex at an adjacent operating station. As coupling/decoupling activities are only allowed at operating stations, the train composition mode is fixed on each traveling trip in the process of operation planning.

- **Rolling stock path:** A rolling stock path consists of a sequence of space–time vertexes that are connected by pull-out arcs, traveling arcs, waiting arcs, turn-around arcs and pull-in arcs in the whole planning time horizon. For convenience, let p denotes a rolling stock path, and $\mathcal{P}^k(k \in \mathcal{K})$ indicates the available path set corresponding to rolling stock unit k . With the space–time path representation, the departure and arrival events at each station for each rolling stock unit can be specified accurately.

For clarity, an illustration of the space–time network is given in Fig. 4, which includes five stations and three depots. Stations s_1 , s_3 and s_5 , having side-tracks for coupling/decoupling activities, are connected with three independent depots, respectively. In this space–time network, the planning time horizon with a duration of 50 min is discretized by a time granularity (i.e., 2 min), and a total of 26 timestamps are involved. We use space–time paths of two rolling stock units to illustrate the operation scheme planning, where rolling stock units k_1 and k_2 are dispatched from depots d_1 and d_2 , respectively. As shown, rolling stock unit k_1 is firstly dispatched to leave depot d_1 using a space–time arc $(d_1, s_1, 0, 2)$. When it arrives at station s_3 , rolling stock unit k_1 will be coupled with k_2 that is dispatched from depot d_2 . Afterwards, this large train composition undertakes the subsequent services, until it returns to station s_3 after turning around at station s_5 . Then, the large composition is decoupled into two rolling stock units again: rolling stock unit k_1 executes the remainder services and then goes back to depot d_1 ; rolling stock unit k_2 goes back to depot d_2 directly.

Typically, both two rolling stock paths will merge into one trajectory in the generated train timetable, in which four trips are involved, i.e., trips r_1 , r_2 , r_3 and r_4 displayed in Fig. 4. As the space–time path of rolling stock unit k_1 overlaps with that of rolling stock unit k_2 partially, the services on the corresponding segments will be executed by the coupled rolling stock units, i.e., the coupling and decoupling activities need take place at the beginning and ending time of the overlapping path. Moreover, the train composition mode on each trip is fixed and unchangeable (trips r_1 and r_4 are undertaken by a single rolling stock unit, and trips r_2 and r_3 are undertaken by two coupled rolling stock units). Practically, this flexible train composition mode can well meet the diverse passenger demand over different space–time regions, through reasonably optimizing the rolling stock schedules.

3.2. Space–time network for passenger flow movements

To characterize the movement of passengers, the passenger flow will be embedded into the space–time network by introducing passenger activity arcs. To keep the consistency with the space–time network of rolling stock, we assume that passengers arrive at each station on each timestamp in $\mathcal{T}$. Then, inflow passengers are input into the metro system at each space–time vertex of stations. The space–time network for passenger flow movements is then denoted by $\mathcal{G}^P = (\mathcal{V}^P, \mathcal{A}^P)$, where $\mathcal{V}^P$ and $\mathcal{A}^P$ represent the sets of passenger-related space–time vertices and arcs. To well characterize the train–passenger relationship, $\mathcal{G}^P$ actually includes part of the space–time network of rolling stock (i.e., all traveling arcs and waiting arcs at each station). Besides, a set of extra space–time vertexes is further constructed to indicate the arrivals of passengers. These arrival vertexes will be connected with the space–time traveling and waiting arcs by introducing passenger waiting arcs, passenger getting-on/off arcs, etc. In addition, a passenger sink vertex is also constructed to characterize flow conversation for outbound passenger flows at each station. Next, we formally introduce space–time arcs in the space–time network for passenger flow movement.

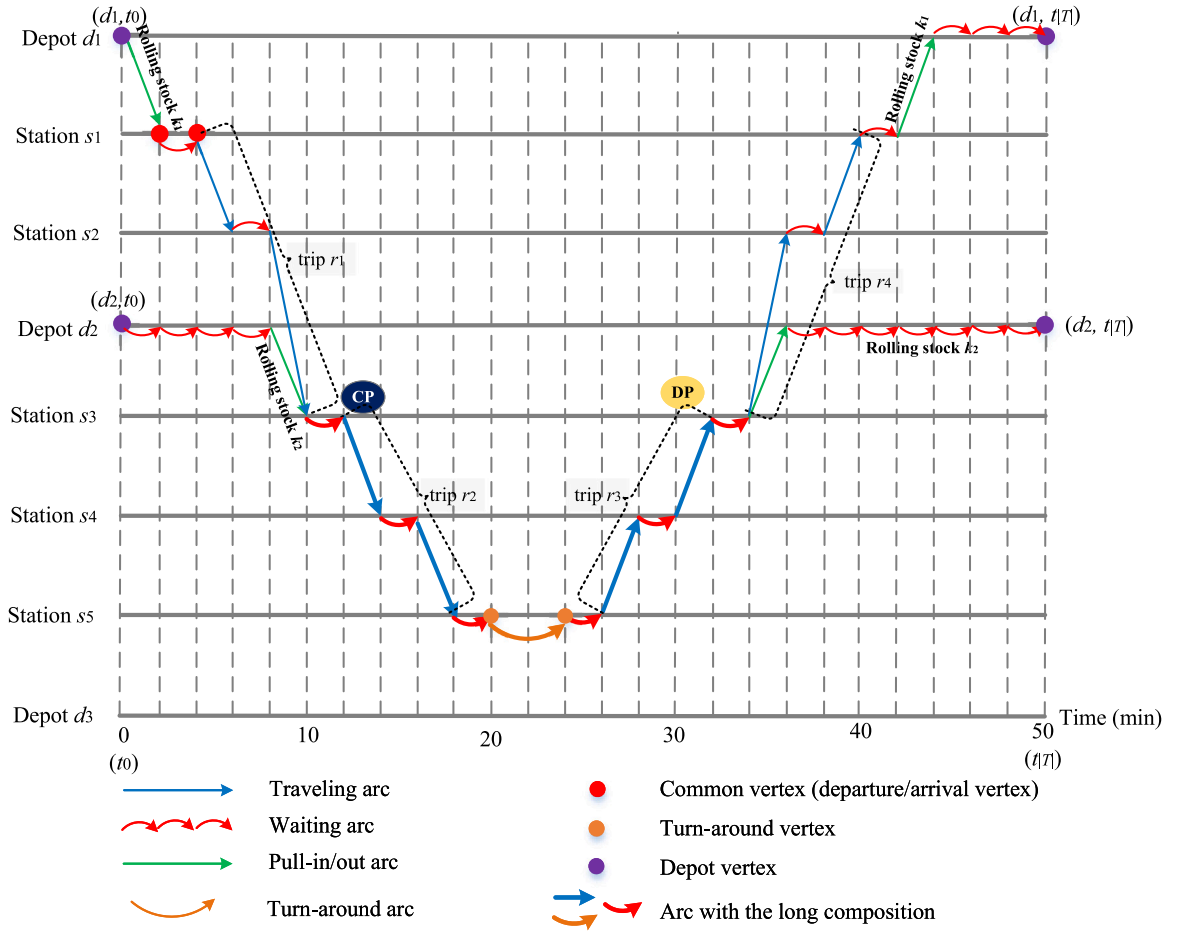


Fig. 4. Arc-based space-time network for the PRSFCMD instance with 3 depots.

• **Passenger arrival arcs:** These arcs indicate the arrival activities of passengers at each space-time arrival vertex, which represent the dynamic inflow passengers at each station. We use notation P_{it}^{in} to represent the volume of inflow passengers at station i on timestamp t .

• **Passenger waiting arcs:** These arcs, which are constructed to represent the waiting state of passengers at each station, exist between each pair of adjacent passenger arrival vertexes. Since all passengers are required to wait at the platform before they get on a train, a variable CP_{it} can be used to denote the volume of cumulative passengers at station i on timestamp t .

• **Passenger getting-on/off arcs:** For each rolling stock traveling arc $a = (i, j, t, t')$, a passenger getting-off arc from vertex (j, t') to the passenger sink vertex is constructed to indicate the volume of alighting passengers at station j . Likewise, a passenger getting-on arc, which indicates the number of passengers boarding a train, is constructed from passenger arrival vertex (i, t) to rolling stock vertex (i, t) . In addition, the total number of passengers getting on a train should not exceed its remainder capacity.

Fig. 5 is given to illustrate the construction of passenger-related space-time network, where only two stations (i.e., i and j) are involved for illustrative purpose, and the trajectories of two rolling stock units are displayed by blue thick lines. To describe the passenger dynamics, a series of passenger arrival vertexes are added to each station on each timestamp. An inflow arc is then introduced to indicate the volume of arrival passengers at each vertex. For instance, a total of 10 passengers arrive at station i on timestamp 1, and 5 passengers arrive at station i on timestamp 2. Moreover, a virtual sink vertex is also constructed at each station to indicate the destination of outflow passengers (i.e., alighting passengers). It is required that the total volume of outflow passengers at all stations should equal to the sum of inbound passengers at all stations during the considered time horizon (i.e., satisfying the passenger flow conservation).

In addition, the passenger waiting arcs are introduced between any two adjacent passenger arrival vertexes in order to compute the accumulated waiting passengers. As shown, the number in each circle denotes the volume of waiting passengers at the current space-time vertex. When a train arrives at the station, a getting-off arc (i.e., the thick red arc) is drawn to show the number of alighting passengers from this train. Likewise, if a train departs from the current station, a getting-on arc (i.e., the thick purple arc) is introduced to indicate the loading process of passengers on this timestamp. For instance, a total of 30 passengers will get off the

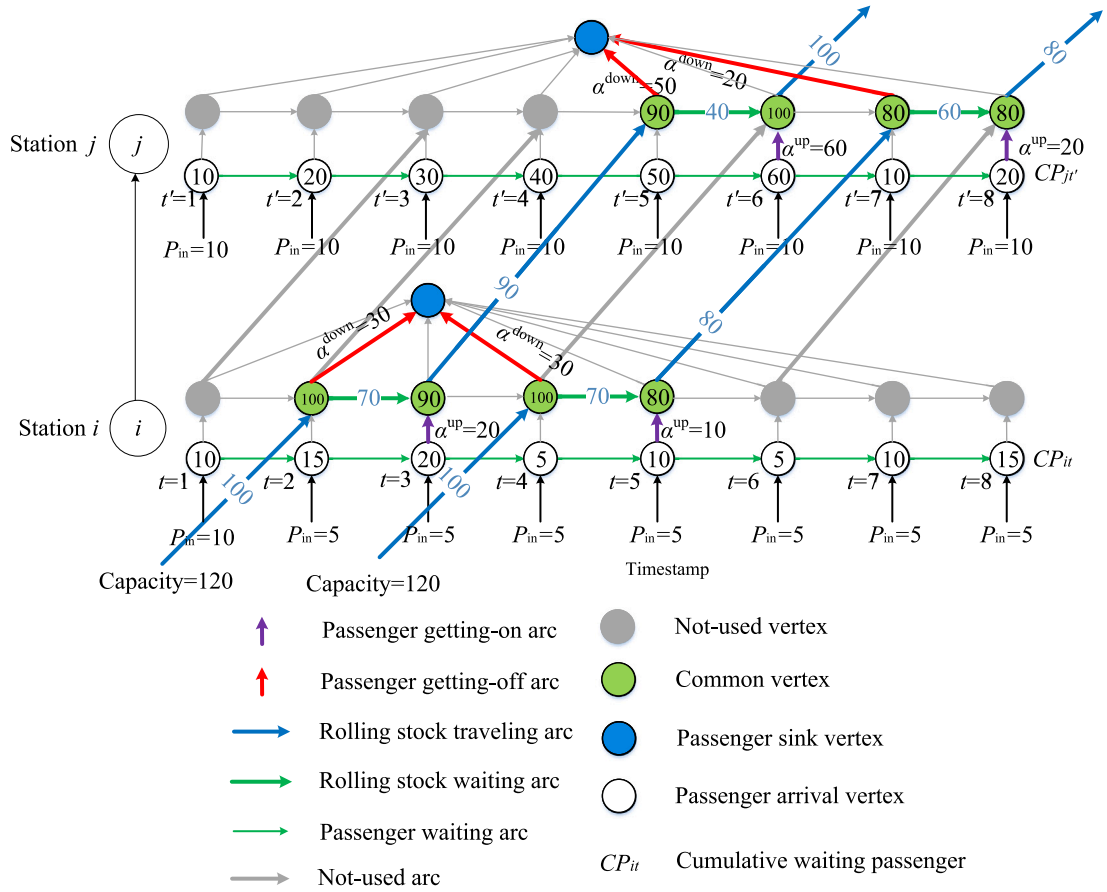


Fig. 5. Space-time network representation for dynamic passenger flow.

train when it arrives at station i on timestamp 2, and a total of 20 passengers will get on the train when it departs from this station on timestamp 3.

With the space-time network representation for rolling stock and passengers, the purpose of this study is to flexibly schedule the rolling stock units that are stored in multiple depots so as to satisfy the passenger demand in the metro system. In this way, it is expected to well match the relationship between the passenger distribution and supplied transport capacity, which might reduce the total operation cost (e.g., with respect to energy consumption, vehicle utilization, etc.) and simultaneously improve the loading occupancy in the operations.

3.3. Assumptions

Without the loss of generality, the following assumptions are firstly given with respect to characteristics of passenger demand and train operations in our considered problem.

Assumption 1. Our study aims to formulate a network-flow-based model for the problem of interest. In data preparation, the time-varying volumes of inflow and outflow passengers at each station will be treated as passenger distribution information. All the passenger-flow-related variables in the formulation are treated as continuous variables for simplicity.

Assumption 2. A normal operation scenario is considered for scheduling rolling stock on the focused metro line, in which the all-stop mode is taken into consideration in the modeling process (i.e., all trains should dwell at each station). In reality, this is a common operation strategy for the real-world applications, e.g., in the Beijing metro.

Assumption 3. The train's arrival and departure activities at a station/depot can only occur on the discretized timestamps. In-vehicle passengers are allowed to get off a train at its arrival vertex, and waiting passengers can get on a train at its departure vertex. Detailed boarding and alighting processes of passengers will be not taken into account for simplicity.

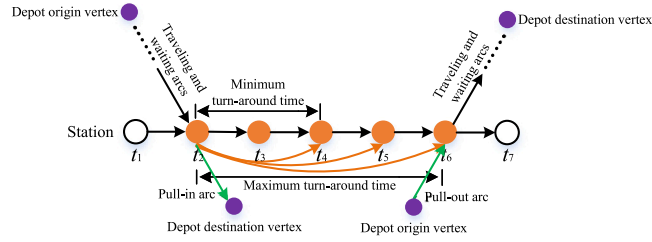


Fig. 6. Rolling stock flow conservation at stations with pull-in/out and turn-around arcs.

Assumption 4. In the operation scheme, coupling and decoupling activities are permitted to be conducted at a few designated operating stations, e.g., at those stations that connect with depots or that have turn-around tracks.

Assumption 5. The maximum dwell time at common stations (except operating stations for possible coupling/decoupling operations) should be less than or equal to the minimum safety headway time in the planning process. This assumption is necessary for keeping coupling/decoupling operations at only operating stations, which can be further concluded based on the following Proposition 4.1.

4. Mathematical formulations

This section aims to develop two optimization models for the problem of interest, i.e., the arc-based model and the path-based model. The detailed modeling process will be elaborated in the following discussion, including notations, parameters, system constraints and the objective function. Besides, the complexity of these two proposed models regarding the decision variables and constraints is also analyzed in detail. For the modeling convenience, the involved indices, sets and parameters used in the formulation are all listed in Table 2.

4.1. Arc-based model in the space–time network

Using the space–time network representation, we firstly formulate an arc-based model for the rolling stock scheduling problem under study, in which the passenger flow is in particular incorporated so as to improve the service level of the generated operation scheme. Three types of decision variables are introduced to develop this model, including: space–time arc selection variables $x_{i,j,t,t'}^k$ indicating the running trajectory of rolling stock in the space–time network, arc passenger flow variables $f_{i,j,t,t'}$ indicating the volume of passengers on each space–time arc, and cumulative passenger flow variables CP_{it} indicating the number of delayed passengers at each station. The feasibility of the joint optimization model can be guaranteed by introducing a set of system constraints, including rolling stock trajectory related constraints (i.e., Eqs. (1)–(4)) and passenger flow related constraints (i.e., Eqs. (5)–(9)), formulated below.

• System Constraints:

(a) Flow balance constraints on space–time vertices for rolling stock

$$\sum_{(i,j,t,t') \in \mathcal{A}_{it}^+} x_{i,j,t,t'}^k - \sum_{(j,i,t',t) \in \mathcal{A}_{it}^-} x_{j,i,t',t}^k = \begin{cases} 1 & (i,t) = (d_{ori}(k), t_0) \\ -1 & (i,t) = (d_{des}(k), t_{|T|}) \\ 0 & \text{otherwise} \end{cases} \quad \forall k \in \mathcal{K} \quad (1)$$

Constraints (1) refer to flow balance constraints for each rolling stock unit k from its depot vertex $(d_{ori}(k), t_0)$ at the beginning of the time horizon to its depot vertex $(d_{des}(k), t_{|T|})$ at the end of the time horizon. For the convenience of the model decomposition and matching relationship between arcs and rolling stock units in the space–time network (Meng and Zhou, 2014; Zhang et al., 2019), such a flow balance constraint is formulated for individual rolling stock unit $k \in \mathcal{K}$ instead of grouping them into multiple rolling stock types. Two sets, i.e., $\mathcal{A}_{it}^+$ and $\mathcal{A}_{it}^-$, denote the outgoing arc set and the incoming arc set of vertex (i,t) in the space–time network of rolling stock. Typically, as the constructed space–time network is acyclic, these constraints will necessarily produce a space–time path for rolling stock unit k in the network. Particularly, based on such flow balance constraints, not only rolling stock flow conservation in traveling and dwelling operations but also pull-in/out and turn-around operations can be well guaranteed without formulating additional circulation constraints in comparison with existing studies (Wang et al., 2018; Zhu and Goverde, 2019), as shown in Fig. 6.

(b) Arc covering constraints

$$\sum_{k \in \mathcal{K}} x_{i,j,t,t'}^k \leq N_{i,j,t,t'}^{\max} \quad \forall (i,j,t,t') \in \mathcal{A}^{RT} \quad (2)$$

Table 2
Definition of involved indices, sets and parameters.

Notation	Definition
Indices	
i, j	Index of physical nodes (stations, depots)
k	Index of rolling stock units
p	Index of rolling stock paths (columns)
(i, j)	Index of physical links
$(i, t), (j, t')$	Index of space-time vertices
$(d_{ori}(k), t_0)$	Index of origin space-time vertex of rolling stock unit k
$(d_{des}(k), t_{ T })$	Index of destination space-time vertex of rolling stock unit k
$a = (i, j, t, t')$	Index of space-time arcs
t, t'	Index of timestamps
Sets	
$\mathcal{I}$	Set of physical nodes
$\mathcal{I}^{OS}$	Set of physical nodes of operating stations with coupling/decoupling operations, $\mathcal{I}^{OS} \subset \mathcal{I}$
$\mathcal{L}$	Set of physical links
$\mathcal{P}^k$	Set of available paths corresponding rolling stock unit k
$\mathcal{T}$	Set of timestamps
$\mathcal{V}^R$	Set of space-time vertices of rolling stock
$\mathcal{A}^R$	Set of space-time arcs of rolling stock
$\mathcal{A}^{R_T}$	Set of traveling arcs of rolling stock, $\mathcal{A}^{R_T} \subset \mathcal{A}^R$
$\mathcal{A}^{R_W}$	Set of waiting arcs of rolling stock, $\mathcal{A}^{R_W} \subset \mathcal{A}^R$
$\mathcal{A}^{OS+}$	Set of outgoing arcs with respect to nodes in $\mathcal{I}^{OS}$, $\mathcal{A}^{OS+} \subset \mathcal{A}^R$
$\mathcal{A}_k^{R_T}$	Set of traveling arcs for rolling stock space-time paths in $\mathcal{P}^k$, $\mathcal{A}_k^{R_T} \subset \mathcal{A}^R$
$\Omega_a / \Omega(i, j, t, t')$	Set of incompatible arcs with respect to $a = (i, j, t, t')$
$\mathcal{K}$	Set of rolling stock units
$\mathcal{V}^P$	Set of space-time vertices of passengers
$\mathcal{A}^P$	Set of space-time arcs of passengers
Parameters	
Cap^k	Capacity of rolling stock unit k (unit: passenger)
p_{it}^{in}	Volume of inflow passengers at space-time vertex (i, t)
p_{it}^{out}	Total volume of outflow passengers during the planning time horizon at station $i \in \mathcal{I}$
C_{kp}^R	Traveling cost for path p with rolling stock unit k
$\bar{C}_{kp}^R$	Reduced cost for rolling stock path p with rolling stock unit k
$N_a^{\text{max}} / N_{i,j,t,t'}^{\text{max}}$	Maximum number of rolling stock units on space-time arc $a = (i, j, t, t')$
$h^A(i, j) / h^D(i, j)$	Minimum arrival/departure headway on link (i, j)
π	Dual variables for involved constraints
$a_{it}^{\text{up}} / a_{it}^{\text{down}}$	Volume of passengers getting on/off trains at station i and timestamp t
γ_p^a	Binary indicator such that $\gamma_p^a = 1$ if rolling stock path p goes through arc a
Decision variables	
$x_{i,j,t,t'}^k$	Binary variables, =1 if space-time arc (i, j, t, t') is selected by rolling stock unit k ; =0 otherwise
y_p	Binary variables, =1 if rolling stock path p is selected; =0 otherwise
$f_{i,j,t,t'} / f_a$	Continuous variables, representing passenger flow volume on space-time arc $a = (i, j, t, t')$
CP_{it}	Continuous variables, representing cumulative passenger flow volume at space-time vertex (i, t)

These constraints refer to the covering constraint for each traveling arc $a = (i, j, t, t') \in \mathcal{A}^{R_T}$ in the space-time network of rolling stock. With the flexible train composition mode, the maximum number of rolling stock units that are allowed to service each space-time arc $a = (i, j, t, t')$ is assumed to be $N_{i,j,t,t'}^{\text{max}}$. For example, $\sum_{k \in \mathcal{K}} x_{i,j,t,t'}^k \leq 2$ means that at most two coupled rolling stock units can cover an arc (i, j, t, t') during the planning time horizon.

(c) *Safety headway constraints*

$$x_{i,j,t,t'}^k + \sum_{(i,j,\tau,\tau') \in \Omega(i,j,t,t')} x_{i,j,\tau,\tau'}^{k'} \leq 1 \quad \forall k' \in \mathcal{K} \setminus \{k\}, k \in \mathcal{K}, (i, j, t, t') \in \mathcal{A}^{R_T} \quad (3)$$

Constraints (3) are safety headway constraints with respect to traveling arcs in the space-time network of rolling stock. For operation safety, the minimum headway $h^A(i, j) / h^D(i, j)$ for the arrival/departure events should be guaranteed between two individual and distinguished rolling stock units on each physical segment (i, j) . To this end, an incompatible set $\Omega(i, j, t, t')$ for $(i, j, t, t') \in \mathcal{A}^{R_T}$ is

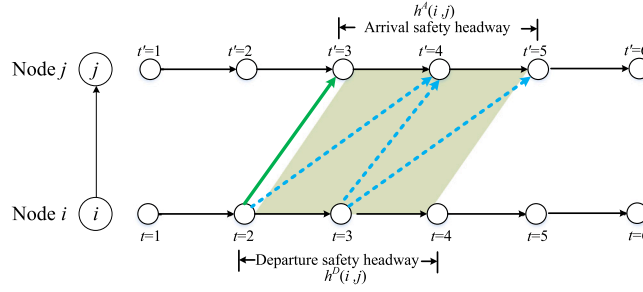


Fig. 7. Illustration for incompatible arc set.

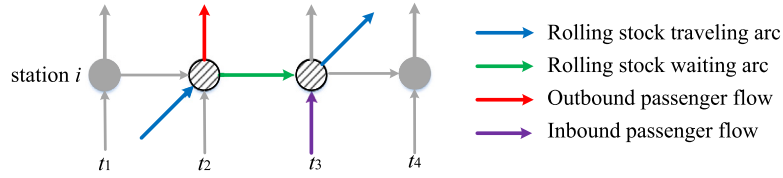


Fig. 8. Illustration for passenger flow conversation on the train.

firstly pre-generated to represent the conflict relationships between different traveling arcs, formulated below.

$$\Omega(i, j, t, t') = \left\{ (i, j, t, t') \in \mathcal{A}^{R_T} \mid 0 < (t - t') < h^D(i, j) \text{ or } 0 < (t' - t) < h^A(i, j) \right\} \quad (4)$$

In the modeling process, all possible conflicting arcs with respect to $a = (i, j, t, t')$ should be collected in set $\Omega(i, j, t, t')$ as the input information. As illustrated in Fig. 7, when safety headway $h^A(i, j)$ and $h^D(i, j)$ are assumed to be 2 min, the incompatible set with respect to arc $(i, j, 2, 3)$ can be represented by

$$\Omega(i, j, 2, 3) = \{(i, j, 2, 4), (i, j, 3, 4), (i, j, 3, 5)\},$$

which includes arcs that overlap (or partially) the shadow area filled with the green color. If the space-time arc $(i, j, 2, 3)$ is used by a rolling stock unit in the planning process, the corresponding constraints (3) will then be activated, and any arc in set $\Omega(i, j, 2, 3)$ might not be selected by other rolling stock units. Note that safety headway constraints can satisfy the flexible train composition mode (see Proposition 4.1). In other words, if a rolling stock unit k occupies arc (i, j, t, t') , this space-time arc can also be serviced by other rolling stock units at the same time.

(d) Passenger flow conversation constraints on trains

$$\sum_{(i, j, t, t') \in \mathcal{A}_{it}^{R_W^+}} f_{i, j, t, t'} - \sum_{(j, i, t', t) \in \mathcal{A}_{it}^{R_T^-}} f_{j, i, t', t} = -\alpha_{it}^{\text{down}} \quad \forall i \in I, t \in \mathcal{T} \quad (5)$$

$$\sum_{(i, j, t, t') \in \mathcal{A}_{it}^{R_T^+}} f_{i, j, t, t'} - \sum_{(j, i, t', t) \in \mathcal{A}_{it}^{R_W^-}} f_{j, i, t', t} = \alpha_{it}^{\text{up}} \quad \forall i \in I, t \in \mathcal{T} \quad (6)$$

With the passenger activity arcs shown in Fig. 5, it is needed to keep the conversation of the passenger flow in the space-time network. We use $\mathcal{A}_{it}^{R_W^+}(\mathcal{A}_{it}^{R_W^-})$ to denote the set of outgoing (incoming) waiting arcs of vertex (i, t) , and use $\mathcal{A}_{it}^{R_T^+}(\mathcal{A}_{it}^{R_T^-})$ to denote the set of outgoing (incoming) traveling arcs of vertex (i, t) in the space-time network for passenger flow movement. In passenger data preparation, we record the time-varying volume of inflow passengers and total outflow passengers at each station. Thus, the flow conversation regarding all the inflow and outflow passengers should be guaranteed. Since passengers can only get off trains at the arrival vertices, the number of outflow passengers $\alpha_{it}^{\text{down}}$ should equal to the difference of flows on incoming traveling arcs and outgoing waiting arcs of (i, t) . Likewise, as passengers can only get on trains at the departure vertices, the flow balance constraints with respect to inbound passenger flow α_{it}^{up} can be formulated in a similar way, as shown in Fig. 8.

(e) Passenger variation constraints at stations

$$CP_{it} = CP_{i, t-1} - \alpha_{it}^{\text{up}} + P_{it}^{\text{in}} \quad \forall i \in I, t \in \mathcal{T} \setminus \{0\} \quad (7)$$

In the scheduling process, we need to characterize the evolution of the passenger flow in order to capture the dynamic variation of passengers inside the metro system (Zhou et al., 2022a). The cumulative passenger flow CP_{it} at vertex (i, t) should equal to the sum of the cumulative passenger flow $CP_{i, t-1}$ at vertex $(i, t-1)$ and the volume of inflow passengers P_{it}^{in} after subtracting the passengers getting on the rolling stock units (i.e., α_{it}^{up}).

(f) Passenger flow conversation at passenger sink vertices

$$\sum_{t \in \mathcal{T}} \alpha_{it}^{\text{down}} = P_i^{\text{out}} \quad \forall i \in \mathcal{I} \quad (8)$$

In this constraint, it is required that the total number of outflow passengers $\alpha_{it}^{\text{down}}$ during the planning time horizon equals to the total outflow passenger P_i^{out} at station $i \in \mathcal{I}$.

(g) Capacity constraints for traveling and waiting arcs

$$\sum_{k \in \mathcal{K}} \text{Cap}^k \cdot x_{i,j,t,t'}^k \geq f_{i,j,t,t'} \quad \forall (i, j, t, t') \in \mathcal{A}^{R_W} \cup \mathcal{A}^{R_T} \quad (9)$$

Constraints (9) represent the capacity constraints for the onboard passengers, which are active for the rolling stock traveling and waiting arcs. Note that, the space-time arcs related to turn-around and pull-in/out operations are not available for passengers. Then, the capacity constraints on these arcs are not taken into consideration. In this set of constraints, we ensure that the number of in-vehicle passengers should not exceed the total capacity of involved rolling stock units on the corresponding space-time traveling or waiting arcs.

(h) Decision variables constraints

All the space-time arc selection decision variables should be binary variables, and all passenger flow volume variables at space-time vertexes and arcs should be positive continuous variables, stated in the following constraints.

$$x_{i,j,t,t'}^k \in \{0, 1\} \quad \forall (i, j, t, t') \in \mathcal{A}^R, k \in \mathcal{K} \quad (10)$$

$$f_{i,j,t,t'} \geq 0 \quad \forall (i, j, t, t') \in \mathcal{A}^P \quad (11)$$

$$CP_{it} \geq 0 \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (12)$$

• **Objective function:**

$$\min F_1 = \alpha \left(\sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} CP_{it} \right) + \beta \left(\sum_{k \in \mathcal{K}} \sum_{(i,j,t,t') \in \mathcal{A}^R} c_{i,j,t,t'} \cdot x_{i,j,t,t'}^k \right) \quad (13)$$

The objective function aims to minimize the total passenger waiting cost at all stations and the total operation cost of all selected arcs in the space-time network of rolling stock. Specifically, the first term in the objective function (13) represents the sum of waiting cost for cumulative passengers at different stations; the second term represents the total operation cost with respect to rolling stock. In addition, parameters α and β are cost coefficients of these two components, respectively.

• **Mathematical model:**

According to Eqs. (1)–(13), the arc-based model (M1) for the PRSFCMD can be formulated as below:

$$\text{M1: } \begin{cases} \min F_1 = \alpha \left(\sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} CP_{it} \right) + \beta \left(\sum_{k \in \mathcal{K}} \sum_{(i,j,t,t') \in \mathcal{A}^R} c_{i,j,t,t'} \cdot x_{i,j,t,t'}^k \right) \\ \text{s.t. Constraints (1)–(3), (5)–(12)} \end{cases} \quad (14)$$

4.2. Path-based model in the space-time network

Note that, a large number of decision variables and constraints regarding arc selections are taken into consideration in the arc-based model. In particular, to find a feasible space-time path for each rolling stock, a variety of constraints need to be formulated to guarantee the flow balance of rolling stock, and the coupling relationships with other constraints might also lead to the solution complexity and then lower the computational efficiency. With this concern, we intend to reformulate this model into a new form that can be handled by a column generation framework, in which the path finding is conducted via a series of shortest path problems. This model is then referred as the path-based model.

Let $\mathcal{P}^k$ be a set of feasible space-time paths for rolling stock unit $k \in \mathcal{K}$ in the space-time network, and denote C_p^R as the cost of path $p, p \in \mathcal{P}^k$. One rolling stock path includes a space-time trajectory starting from one depot origin vertex $(d_{ori}(k), t_0)$ and ending to one depot destination vertex $(d_{des}(k), t_{|T|})$ during the considered planning time horizon, with a series of pull-in/out arcs, traveling arcs, waiting arcs and turn-around arcs in the space-time network $\mathcal{G}^R = (\mathcal{V}^R, \mathcal{A}^R)$. Accordingly, we could then obtain the feasible solution of the proposed problem based on a combination of the rolling stock paths. In this formulation, the path selection variables y_p are newly introduced. The arc-based model can be transformed into a path-based model with a set of constraints regarding rolling stock path assignment, arc covering, safety headway and passenger flow assignment as given below. For simplicity, the space-time arc index (i, j, t, t') will be abbreviated to a symbol a in the modeling process. Considering that all rolling stock units need to go through a space-time path from virtual origin to destination vertexes (no matter path for actual operations or waiting at depots), a constraint with the equality in Eq. (15) is involved in the rolling stock path assignment constraints.

• **Constraints:**

(a) Rolling stock path assignment constraints

$$\sum_{p \in \mathcal{P}^k} y_p = 1 \quad \forall k \in \mathcal{K} \quad (15)$$

(b) Arc covering constraints

$$\sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}^k} \gamma_p^a y_p \leq N_a^{\max} \quad \forall a \in \mathcal{A}^{R_T} \quad (16)$$

(c) Safety headway constraints

$$\sum_{p \in \mathcal{P}^k} \gamma_p^a y_p + \sum_{p \in \mathcal{P}^{k'}} \sum_{a' \in \Omega_a} \gamma_p^{a'} y_p \leq 1 \quad \forall k' \in \mathcal{K} \setminus \{k\}, k \in \mathcal{K}, a \in \mathcal{A}^{R_T} \quad (17)$$

(d) Passenger flow related constraints

$$\sum_{a \in \mathcal{A}_{it}^{R_W^+}} f_a - \sum_{a \in \mathcal{A}_{it}^{R_T^-}} f_a = -\alpha_{it}^{\text{down}} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (18)$$

$$\sum_{a \in \mathcal{A}_{it}^{R_T^+}} f_a - \sum_{a \in \mathcal{A}_{it}^{R_W^-}} f_a = \alpha_{it}^{\text{up}} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (19)$$

$$CP_{it} = CP_{i,t-1} - \alpha_{it}^{\text{up}} + P_{it}^{\text{in}} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \setminus \{0\} \quad (20)$$

$$\sum_{i \in \mathcal{I}} \alpha_{it}^{\text{down}} = P_i^{\text{out}} \quad \forall i \in \mathcal{I} \quad (21)$$

$$\sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}^k} \text{Cap}^k \cdot \gamma_p^a \cdot y_p \geq f_a \quad \forall a \in \mathcal{A}^{R_T} \cup \mathcal{A}^{R_W} \quad (22)$$

$$y_p \in \{0, 1\} \quad \forall p \in \mathcal{P}_k, k \in \mathcal{K} \quad (23)$$

$$f_{i,j,t,t'} \geq 0 \quad \forall (i, j, t, t') \in \mathcal{A}^P \quad (24)$$

$$CP_{it} \geq 0 \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (25)$$

• **Objective function:**

$$\min F_2 = \alpha \left(\sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} CP_{it} \right) + \beta \left(\sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}^k} C_{kp}^R \cdot y_p \right) \quad (26)$$

In constraints (15), we ensure each rolling stock unit $k \in \mathcal{K}$ to be assigned to only one rolling stock path $p \in \mathcal{P}^k$. Constraints (16) and (17), respectively, refer to the arc covering constraints and safety headway constraints in the rolling stock scheduling process. To characterize the dynamic evolution of the passenger flow in the metro system, we also equivalently derive a series of passenger flow related constraints in the space-time network for passenger flow movement, including passenger flow conversation constraints on trains (Eqs. (18)–(19)), passenger variation constraints at stations (Eq. (20)), passenger flow conversation constraints at the passenger sink vertices (Eq. (21)) and capacity constraints for traveling and waiting arcs (Eq. (22)). The objective function (Eq. (26)) is also an equivalent reformulation of that in the arc-based model.

• **Mathematical model:**

According to Eqs. (15)–(26), the path-based model (M2) for the PRSFCMD can be formulated as follows:

$$\text{M2:} \begin{cases} \min F_2 = \alpha \left(\sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} CP_{it} \right) + \beta \left(\sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}^k} C_{kp}^R \cdot y_p \right) \\ \text{s.t. Constraints (15)–(25)} \end{cases} \quad (27)$$

4.3. Some discussions of model properties

Next, we are interested in discussing some properties of the proposed models, aiming to highlight some specific characteristics of the flexible train composition mode.

Proposition 4.1. *In the proposed models, arc covering constraints and safety headway constraints necessarily satisfy operation conditions of the flexible train composition mode, i.e., (1) the coupling/decoupling operations are not allowed at common stations (except operating stations); (2) more than one rolling stock unit can cover a space-time arc under the safety headway requirement.*

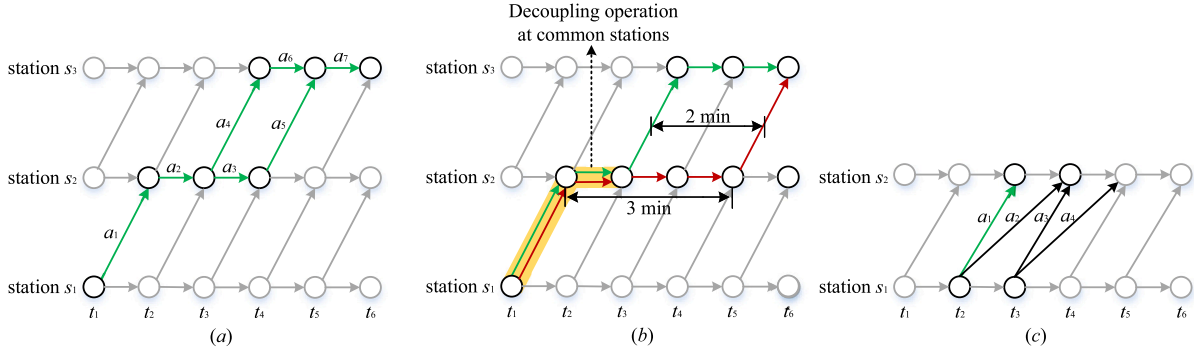


Fig. 9. Illustration for the proposition.

Proof. (1) We here prove that the decoupling operations are not permitted at common stations, and the proof for coupling operations can be obtained similarly. For clarity, we use Fig. 9(a) as an illustration. Without the loss of generality, we suppose the minimum headway time is 2 min, and at most two rolling stock units are allowed to service one arc, i.e., $N_a^{\max} = 2, \forall a \in \mathcal{A}^{RT}$. As shown, station s_2 is a common station, and two coupled rolling stock units arrive at station s_2 at timestamp t_2 by servicing arc a_1 , i.e., $x_{a_1}^{k_1} = x_{a_1}^{k_2} = 1$. Note that the minimum and maximum dwell time durations are 1 min and 2 min, respectively (satisfying Assumption 5). In the scheduling process, only arcs a_1, a_2, a_3, a_4 and a_5 (i.e., green arcs in Fig. 9(a)) can be used for these rolling stock units in the following journey. According to the flow balance constraints related to rolling stock, we have the following equations at space-time vertices $(s_2, t_2), (s_2, t_3)$ and (s_2, t_4) :

$$x_{a_1}^k = x_{a_2}^k, x_{a_2}^k = x_{a_3}^k + x_{a_4}^k, x_{a_3}^k = x_{a_5}^k, \quad \forall k \in \{k_1, k_2\}.$$

In addition, according to the headway constraints, we have

$$x_{a_4}^{k_1} + x_{a_5}^{k_2} \leq 1 \text{ and } x_{a_4}^{k_2} + x_{a_5}^{k_1} \leq 1.$$

By solving this set of inequalities, we obtain $x_{a_4}^{k_1} = x_{a_4}^{k_2} = 1$ or $x_{a_5}^{k_1} = x_{a_5}^{k_2} = 1$, which means that the decoupling operation might not be performed at station s_2 . The first condition is thus guaranteed.

On the other hand, the decoupling operations with such proposition might not be avoided without Assumption 5. For example with a counter case shown in Fig. 9(b), the minimum headway time and maximum dwell time are set to 2 min and 3 min respectively, which violates Assumption 5. By solving inequalities of flow balance and headway constraints, we can obtain a feasible solution with decoupling operations at common station s_2 , which typically conflicts with real-world operations.

(2) In the following, we prove that more than one rolling stock unit can cover a space-time arc under the safety headway requirement. For simplicity, we still set the minimum safe headway time as 2 min. As illustrated in Fig. 9(c), the incompatible set with respect to arc a_1 can be represented by $\Omega(a_1) = \{a_2, a_3, a_4\}$. Then, arc covering constraints and safety headway constraints related to arc a_1 can be formulated in Eq. (28).

$$\begin{cases} \sum_{k \in \mathcal{K}} x_{a_1}^k \leq N_a^{\max} \\ x_{a_1}^k + x_{a_2}^{k'} + x_{a_3}^{k'} + x_{a_4}^{k'} \leq 1 \quad \forall k' \in \mathcal{K} \setminus \{k\}, k \in \mathcal{K} \end{cases} \quad (28)$$

In this equation, if arc a_1 is serviced by a rolling stock unit k_0 , i.e., $x_{a_1}^{k_0} = 1$, we then have

$$\sum_{k' \in \mathcal{K} \setminus \{k_0\}} x_{a_1}^{k'} \leq N_a^{\max} - 1; \text{ and } x_{a_2}^{k'} + x_{a_3}^{k'} + x_{a_4}^{k'} = 0 \quad \forall k' \in \mathcal{K} \setminus \{k_0\}.$$

This means that no other rolling stock units can utilize space-time arcs in set $\Omega(a_1) = \{a_2, a_3, a_4\}$, while at most other $N_a^{\max} - 1$ rolling stock units in $\mathcal{K} \setminus \{k_0\}$ can be coupled with k_0 to service arc a_1 simultaneously. Then, the second condition is guaranteed. $\square$

Proposition 4.2. In the path-based model, it can be found that a huge number of arc covering constraints (16) and safety headway constraints (17) should still be handled in the space-time network, which would lower the solution efficiency. Then, the following simplified constraints (29) and (30) could be adopted dynamically to reduce the number of arc-based constraints after generating specific path sets.

Proof. (1) Denote $\mathcal{I}^{OS}$ as the set of operating stations on the considered metro line. Then, $\mathcal{A}^{OS+}$ is used to denote the set of space-time outgoing arcs originating from the stations in $\mathcal{I}^{OS}$. In practical operations, the train composition mode should be fixed on each trip, and coupling/decoupling activities are only allowed at operating stations. Thus, the arc covering constraints (16) can be simplified as: restricting the number of rolling stock units only on arcs in set $\mathcal{A}^{OS+}$, shown in Eq. (29).

$$\sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}^k} \gamma_p^a y_p \leq N_a^{\max} \quad \forall a \in \mathcal{A}^{OS+} \quad (29)$$

Table 3
Number of decision variables and constraints in different formulations.

Variables or constraints	Arc-based model	Path-based model
Space–time arc selection variables $x_{i,j,t,t'}^k$	$ \mathcal{K} \mathcal{A}^R $	–
Path selection variables y_p	–	$\sum_{k \in \mathcal{K}} \mathcal{P}^k $
Arc passenger flow variables $f_{i,j,t,t'}/f_a$	$ \mathcal{A}^P $	$ \mathcal{A}^P $
Cumulative passenger flow variables CP_{it}	$ \mathcal{I} \mathcal{T} $	$ \mathcal{I} \mathcal{T} $
Flow balance constraints on space–time vertices for rolling stock	$ \mathcal{K} \mathcal{V}^R $	–
Rolling stock path assignment constraints	–	$ \mathcal{K} $
Arc covering constraints	$ \mathcal{A}^{R_T} $	$ \mathcal{A}^{SO+} $
Safety headway constraints	$ \mathcal{K} (\mathcal{K} - 1) \mathcal{A}^{R_T} $	$\sum_{k \in \mathcal{K}} (\mathcal{K} - 1) \mathcal{A}_k^{R_T} $
Passenger flow conversation constraints on trains	$2 \mathcal{I} \mathcal{T} $	$2 \mathcal{I} \mathcal{T} $
Passenger variation constraints at stations	$ \mathcal{I} \mathcal{T} $	$ \mathcal{I} \mathcal{T} $
Passenger flow conversation constraints at passenger sink vertices	$ \mathcal{I} $	$ \mathcal{I} $
Capacity constraints for traveling and waiting arcs	$ \mathcal{A}^{R_w} + \mathcal{A}^{R_T} $	$ \mathcal{A}^{R_w} + \mathcal{A}^{R_T} $

(2) On the other hands, safety headway constraints (17) can also be generated dynamically in order to reduce the scale of model M2 with deterministic column pools. When the column pool $\mathcal{P}^k$ with respect to rolling stock unit k is given, we denote $\mathcal{A}_k^{R_T}$ as the set of traveling arcs for space–time paths in $\mathcal{P}^k$. It is true that only arcs in set $\mathcal{A}_k^{R_T}$ may conflict with other arcs attached to other rolling stock units. Therefore, constraints (17) can be rewritten as constraints (30) with the given column pool. $\square$

$$\sum_{p \in \mathcal{P}^k} \gamma_p^a y_p + \sum_{p \in \mathcal{P}^{k'}} \sum_{a' \in \Omega_a} \gamma_p^{a'} y_p \leq 1 \quad \forall k' \in \mathcal{K} \setminus \{k\}, a \in \mathcal{A}_k^{R_T}, k \in \mathcal{K} \quad (30)$$

4.4. Complexity analyses

Note that the problem scale and solution efficiency might be different for these two formulation methods. To specify the complexity of the proposed models, we next display the scale of the proposed two formulations in Table 3, in which the symbol $|\cdot|$ denotes the cardinality of an involved set.

We observe from this table that, the complexity of the arc-based model is closely related to the scales of the involved network and rolling stock, including the number of space–time vertexes, the number of space–time arcs, the number of rolling stock units, etc. As complex coupling relationships exist between different constraints, it is actually difficult for us to design a customized algorithm for this model, and an optimization solver will be adopted to solve it in our numerical experiments. In comparison, in addition of the network scale, the complexity of the path-based model is also dependent on the number of potential space–time paths. We note that there possibly exists a huge number of paths in the space–time network for each rolling stock unit. To find a high-quality solution within an acceptable computing time, we next aim to design an effective column generation framework to solve the path-based model in the following section.

5. Solution algorithm

In this section, we aim to design a branch-and-price (B&P) algorithm to solve the path-based model formulated above, which is an efficient solution approach to solve the large-scale discrete optimization problems by a combination of column generation and branch-and-bound (B&B) algorithms. A general description of the B&P procedure along with the overall framework of the algorithm is shown in Fig. 10. At the beginning of the search process (i.e., origin node in B&B), a set of feasible columns is initially generated by the dynamic programming (DP) algorithm (see Section 5.2) with the initial volume of inflow passengers. Let $\tilde{\mathcal{P}}^k, \forall k \in \mathcal{K}$ be the initially generated column pool composed of space–time paths for different rolling stock units. Then, the column generation approach is applied to solving the LP relaxation of the path-based model (i.e., M2-LP). Specifically, the initial column pool is firstly taken into account for the formulation of the restricted master problem (RMP) (i.e., the RMP should be feasible with the initial column pool). After solving the RMP by an optimization solver (e.g., CPLEX or Gurobi), dual variables π of constraints are used to construct pricing sub-problems (i.e., $\text{SPR}_k, \forall k \in \mathcal{K}$) to find new promising columns, in order to improve the quality of the solution. If the minimum reduced costs for all pricing sub-problems solved by DP are greater than 0, we can ensure that an optimal solution is found for the M2-LP. Otherwise, the generated columns will be inserted into the column pool for subsequent iterations. This solution process will be iterated until no more columns with negative reduced costs can be generated.

If the generated optimal solution of M2-LP is integer-type, it can be said that we find the optimal solution to the path-based model. Otherwise, the B&B algorithm will be embedded in the searching process to obtain integer solutions. In detail, a fractional column selection solution (i.e., y_p) that is nearest to 0.5 (or a predefined parameter) will be chosen to round to 0 and 1, producing two different search nodes. On each node, the column generation approach is also used to solve the LP relaxation of the path-based model with the given branching conditions. The above search goes on iteratively along with the bounding and pruning processes until no search nodes need to be processed, and the best encountered feasible solution is outputted as a near-optimal solution to the path-based model. Section 5.3 can be referred for a more detailed description of B&B strategies. In addition to the breadth-first-search (BFS) method for generating the B&B search tree, the depth-first-search (DFS) approach can also be used to generate UB solutions in parallel and efficiently tighten the bounding process with fast-generated feasible solutions.

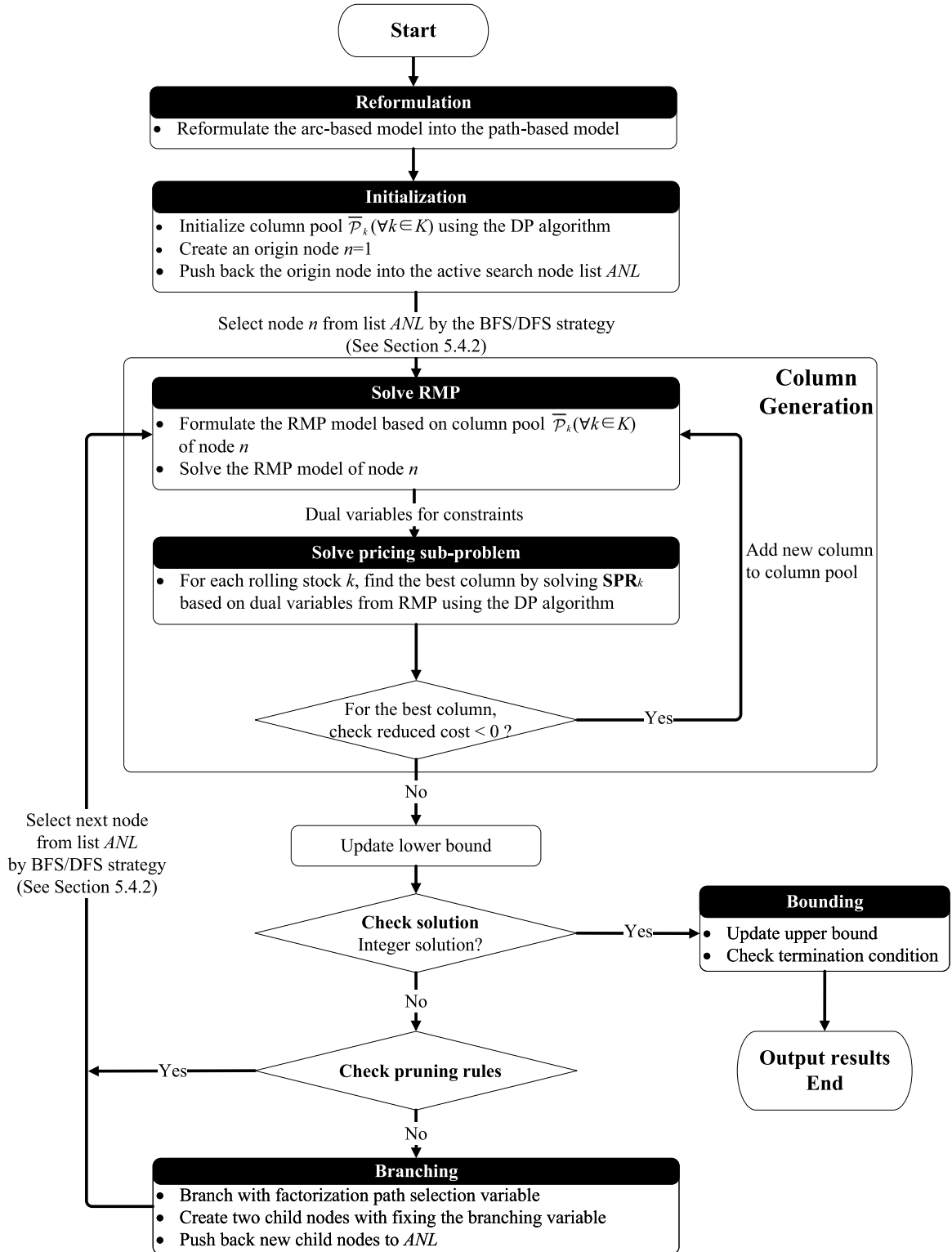


Fig. 10. Overall solution framework in our study.

5.1. Restricted master problem and pricing sub-problem

As stated above, the column generation approach is used to solve the LP relaxation of the path-based model. As the number of potential paths for each rolling stock unit is enormous, the restricted master problem firstly considers an initial column pool, denoted by $\bar{\mathcal{P}}^k, \forall k \in \mathcal{K}$, in the LP relaxation model. For clarity, we formulate the RMP as follows.

$$\text{RMP:} \left\{ \begin{array}{ll} \min & F_3 = \alpha \left(\sum_{i \in \mathcal{I}} \sum_{t \in \mathcal{T}} C P_{it} \right) + \beta \left(\sum_{k \in \mathcal{K}} \sum_{p \in \bar{\mathcal{P}}^k} C_{kp}^R \cdot y_p \right) \\ \text{s.t.} & \sum_{p \in \bar{\mathcal{P}}^k} y_p = 1 \quad \forall k \in \mathcal{K} \quad (a) \\ & \sum_{k \in \mathcal{K}} \sum_{p \in \bar{\mathcal{P}}^k} \gamma_p^a y_p \leq N_a^{\max} \quad \forall a \in \mathcal{A}^{OS+} \quad (b) \\ & \sum_{p \in \bar{\mathcal{P}}^k} \gamma_p^a y_p + \sum_{a' \in \Omega_a} \sum_{p \in \bar{\mathcal{P}}^{k'}} \gamma_{p'}^{a'} y_{p'} \leq 1 \quad \forall k' \in \mathcal{K} \setminus \{k\}, a \in \mathcal{A}_k^{RT}, k \in \mathcal{K} \quad (c) \\ & - \sum_{k \in \mathcal{K}} \sum_{p \in \bar{\mathcal{P}}^k} \text{Cap} \cdot \gamma_p^a \cdot y_p \leq -f_a \quad \forall a \in \mathcal{A}^{RW} \cup \mathcal{A}^{RT} \quad (d) \\ & \text{Constraints (18) – (21), (24) – (25)} \\ & y_p \in [0, 1] \quad \forall p \in \bar{\mathcal{P}}^k, k \in \mathcal{K} \end{array} \right. \quad (31)$$

In this linear programming model, we denote the dual variables corresponding to constraints (31)a–(31)d by π^1 – π^4 respectively, i.e., $\pi^1 = \{\pi_k^1\}_{k \in \mathcal{K}}, \pi^2 = \{\pi_a^2\}_{a \in \mathcal{A}^{OS+}}, \pi^3 = \{\pi_{kk'a}^3\}_{k' \in \mathcal{K} \setminus \{k\}, a \in \mathcal{A}_k^{RT}, k \in \mathcal{K}}, \pi^4 = \{\pi_a^4\}_{a \in \{\mathcal{A}^{RW} \cup \mathcal{A}^{RT}\}}$. Thus, the reduced cost for each path p of rolling stock unit k can be calculated by Eq. (32).

$$\bar{C}_{kp}^R = C_p^R - \pi_k^1 - \sum_{a \in \mathcal{A}^{OS+}} \gamma_p^a \pi_a^2 - \sum_{a \in \mathcal{A}_k^{RT}} \sum_{k' \in \mathcal{K} \setminus \{k\}} \left(\gamma_p^a \pi_{kk'a}^3 + \sum_{a' \in \Omega_a} \gamma_{p'}^{a'} \pi_{kk'a}^3 \right) - \sum_{a \in \{\mathcal{A}^{RW} \cup \mathcal{A}^{RT}\}} \left(-\text{Cap}^k \cdot \gamma_p^a \right) \pi_a^4 \quad \forall p \in \bar{\mathcal{P}}^k, k \in \mathcal{K} \quad (32)$$

The aim of the pricing sub-problem is to find promising space–time paths for rolling stock, which might potentially reduce the objective value of the RMP. Thus, the pricing sub-problem for rolling stock unit $k \in \mathcal{K}$ becomes a problem of finding a space–time path $p \in \bar{\mathcal{P}}^k$ with the minimum reduced cost in the constructed space–time network, as shown in Eq. (33).

$$(\text{SPR}_k) \quad \min_{p \in \bar{\mathcal{P}}^k} \bar{C}_{kp}^R, \quad \forall k \in \mathcal{K} \quad (33)$$

Thus, only the flow balance constraints should be imposed to keep the feasibility of the generated space–time path for rolling stock. As the pricing sub-problems are shortest path problems with flow balance constraints, they can be solved efficiently by the DP in parallel. For clarity, for SPR_k , the arc-based objective function and flow balance constraints can be formulated in Eq. (34), in which a space–time path p is represented by a set of space–time arc selection variables $\{x_{i,j,t,t'}^k\}$ satisfying constraints (34)a for rolling stock unit k .

$$\text{SPR}_k : \left\{ \begin{array}{ll} \min & \sum_{(i,j,t,t') \in \mathcal{A}^R} c_{i,j,t,t'} x_{i,j,t,t'}^k - \sum_{(i,j,t,t') \in \mathcal{A}^{OS+}} \pi_{i,j,t,t'}^2 x_{i,j,t,t'}^k + \sum_{(i,j,t,t') \in \{\mathcal{A}^{RW} \cup \mathcal{A}^{RT}\}} \text{Cap}^k \cdot \pi_{i,j,t,t'}^4 x_{i,j,t,t'}^k \\ & - \sum_{(i,j,t,t') \in \mathcal{A}_k^{RT}} \sum_{k' \in \mathcal{K} \setminus \{k\}} \left(\pi_{k,k',i,j,t,t'}^3 x_{i,j,t,t'}^k + \sum_{(i',j',\tau,\tau') \in \Omega_{i,j,t,t'}} \pi_{k,k',i',j',\tau,\tau'}^3 x_{i',j',\tau,\tau'}^k \right) - \pi_k^1 \\ \text{s.t.} & \sum_{(j,t') \in \mathcal{A}_{it}^+} x_{i,j,t,t'}^k - \sum_{(j,t') \in \mathcal{A}_{it}^-} x_{j,i,t,t'}^k = \begin{cases} 1 & (i,t) = (d_{ori}(k), t_0) \\ -1 & (i,t) = (d_{des}(k), t_{|T|}) \\ 0 & \text{otherwise} \end{cases} \quad (a) \\ & x_{i,j,t,t'}^k \in \{0, 1\} \quad \forall (i,j,t,t') \in \mathcal{A}^R \end{array} \right. \quad (34)$$

We denote the shortest path by p^* after solving model SPR_k . If $\bar{C}_{kp^*}^R \geq 0$, there is no column with the negative reduced cost, and the column generation algorithm terminates. Otherwise, the optimal solution to SPR_k becomes a promising column, and it is added to the column pool $\bar{\mathcal{P}}^k$ in the next iteration.

In addition, we mention that the specific characteristics with respect to the dual variables and the reduced cost might give potential directions for designing an efficient algorithm in solving these sub-problems. Specifically, based on the expression of the reduced cost objective function in Eq. (32), we can observe that a well-generated column should execute less long compositions, have fewer conflicts with other columns, and carry residual passengers as much as possible based on the current solution of RMP.

5.2. Dynamic programming (DP)

5.2.1. DP for generating the initial column pool

To our knowledge, the number of iterations required in the column generation procedure at a search node depends on the structure of the initial column pool. The generation process of an initial column can be viewed as the shortest-path-finding process in the constructed space–time network of rolling stock, which might be solved by a customized DP algorithm efficiently (Yao et al., 2019; Zhang et al., 2019; Wang et al., 2023). By preliminary experiments, we find that a good initial column pool should contain diverse and beneficial columns with less conflicts, and meanwhile might satisfy the passenger demand as much as possible. Therefore, two rules could be embedded into the DP algorithm to generate the initial column pool: (a) no overlapping space–time resource occupation (i.e., no conflicts) could occur between any two rolling stock columns in the initial column pool; (b) with the inflow passengers P_{it}^{in} , the total volume of carried passengers by the rolling stock unit along the time horizon is used as the label in the searching process. Therefore, at each space–time vertexes (i, t) , a passenger-oriented label PAS_{it} is used to represent the total volume of carried passengers from depot $d_{ori}(k)$ on timestamp t_0 for rolling stock unit k .

Then, the pseudo-code of the initial column pool generation procedure is displayed in Algorithm 1. In the label updating process at a space–time vertex (i, t) for a rolling stock unit, we consider the total passenger demand arriving in time interval $[t - \Delta t, t]$, where Δt is a pre-given parameter for estimating the passenger demand for the current rolling stock unit. For instance, a new label $PAS_{jt'}$ at station j on timestamp t' is calculated based on the label PAS_{it} of previous vertex (i, t) if arc (i, j, t, t') exists and on the total passenger demand arriving at station j in time interval $[t' - \Delta t, t']$. After executing the DP for rolling stock unit k , we can select a total of M largest labels at the space–time vertexes of depot $d_{des}(k)$, and then use the trace-back method to find M space–time paths as the initial columns for this rolling stock unit, where M is a pre-given threshold for the number of initial generated columns for a rolling stock unit. Moreover, to reduce conflicts between columns with respect to different rolling stock units, the space–time resources with respect to the generated columns in $\bar{P}^k$ will be marked so that any subsequent rolling stock units avoid to occupy these resources. Through this operation, we can generate an initial column pool for the LP relaxation of the path-based model, which might warmly start the column generation algorithm in the solution process.

Algorithm 1 Dynamic programming for generating initial column pool

Input:

1: The space–time network $\mathcal{G}^R = (\mathcal{V}^R, \mathcal{A}^R)$ of rolling stock; the space–time network $\mathcal{G}^P = (\mathcal{V}^P, \mathcal{A}^P)$ of passengers; the planning time horizon $\mathcal{T}$; the time-dependent inbound passenger flow volume P_{it}^{in} on timestamp t at station i ; the maximum number M of generated initial columns for one rolling stock unit.

Output:

2: The initial column pool $\bar{P}^k, k \in \mathcal{K}$, which contains diverse and beneficial columns with less conflicts, and satisfies the passenger demand as much as possible.
3: **for** each rolling stock unit k **do**
4: **Step 1 (Initialization):**
5: 1. Create an origin label $PAS_{d_{ori}(k), t_0} = 0$ at the depot vertex $(d_{ori}(k), t_0)$.
6: 2. Create labels at other vertices, i.e., $PAS_{it} = -\infty, (i, t) \in \mathcal{V}^R \setminus \{(d_{ori}(k), t_0)\}$.
7: 3. Initialize label predecessors $pred(PAS_{it}) = \text{None}, i \in \mathcal{I}, t \in \mathcal{T}$.
8: 4. Initialize column pool $\bar{P}^k \leftarrow \emptyset$.
9: **Step 2 (Updating label dynamically):**
10: **for** each timestamp $t \in \mathcal{T}$, each node $i \in \mathcal{I}$ **do**
11: **for** each successor node j of node i **do**
12: **for** traveling time TT in the travel time range of link (i, j) **do**
13: 1. Calculate the arrival time t' at node j : $t' = t + TT$.
14: 2. Choose the current label PAS_{it} of space–time vertex (i, t) .
15: 3. Create a new label $PAS'_{jt'}$ on space–time vertex (j, t') from label PAS_{it} .
16: $PAS'_{jt'} = PAS_{it} + \sum_{\tau \in [t' - \Delta t, t']} P_{j\tau}$
17: **if** $PAS'_{jt'} \geq PAS_{jt'}$ **then**
18: 1. Update label $PAS_{jt'} \leftarrow PAS'_{jt'}$.
19: 2. Update label predecessor $pred(PAS_{jt'}) = (i, t)$.
20: **Step 3 (Backtrace):**
21: For rolling stock unit k , based on the label predecessors $pred(PAS_{it})$ and labels PAS_{it} , select a total of M largest labels at the space–time vertexes of depot $d_{des}(k)$, and then trace back to find M space–time paths as the initial column pool $\bar{P}^k$ for this rolling stock unit.
22: **Step 4 (Updating space–time network):**
23: Mark the space–time occupation resources of rolling stock column in path set $\bar{P}^k$, and the marked resource avoids to be occupied by any generated initial columns for other subsequent rolling stock units again.

5.2.2. DP for solving pricing sub-problems

When the pricing problem is decomposed into a sequence of sub-problems related to each rolling stock unit, the sub-problem SPR_k for rolling stock unit $k \in \mathcal{K}$ can be viewed as a time-dependent shortest path problem in the space–time network of rolling stock, which can also be solved by a DP algorithm efficiently (Algorithm 2) in parallel for rolling stock units. Compared with the DP for generating the initial column pool, the total reduced cost RC_{it} is regarded as the label to represent the reduced cost from depot vertex $(d_{des}(k), t_0)$ to current vertex (i, t) with respect to rolling stock unit k . Therefore, in the label updating process, a new label $RC'_{jt'}$ at vertex (j, t') can be calculated based on label RC_{it} , traveling costs and dual values of arc (i, j, t, t') if arc (i, j, t, t') exists, as shown in Eq. (35).

$$RC'_{jt'} = RC_{it} + c_{i,j,t,t'} - \pi_{i,j,t,t'}^2 - \sum_{k' \in \mathcal{K} \setminus \{k\}} \left(\pi_{k,k',i,j,t,t'}^3 + \sum_{(i',j',\tau,\tau') \in \Omega_{i,j,t,t'}} \pi_{k,k',i',j',\tau,\tau'}^3 \right) + \text{Cap}^k \cdot \pi_{i,j,t,t'}^4 \quad \forall (i, j, t, t') \in \mathcal{A}^R, k \in \mathcal{K} \quad (35)$$

Although the goal of each sub-problem is to minimize the total reduced cost of the space–time path, in the searching process, we also need to record the total volume of carried passengers and total operation cost (including traveling, waiting and turn-around costs) on sub-paths from depot vertex $(d_{ori}(k), t_0)$ to current vertex (i, t) , respectively, denoted by C_{it} and PAS_{it} . After performing the DP, a back-tracing procedure is used to find the shortest path based on the minimum negative label (i.e., reduced cost) at the vertex with the arrival event at its destination depot. With three recorded labels, we give the following preference relationship. For two paths from depot origin vertex to vertex with the arrival event at the destination depot, we will keep the path with the least total reduced cost; if these two paths have the same reduced cost, we choose to keep the path with the maximum total volume of carried passengers; if the first two labels are the same, we will keep the path with the least total operation cost. By this operation, we actually define a lexicographical order relationship for all the involved paths. At the same time, a column checking step is processed to exclude the forbidden columns (i.e., column p with $y_p = 0$) based on the branching condition of the current node. Specifically, if the currently selected path (e.g., p_1) happens to be forbidden based on the branching rule (i.e., $y_{p_1} = 0$) unfortunately, then an alternative label at the arrival event vertex of the destination depot $d_{des}(k)$ will be selected with the second minimum reduced cost (still satisfying < 0). Then, another space–time path is back-traced as the new column without violating the branching condition. Finally, this path column will be inserted into the column pool for the subsequent iterations. We note that we simplify and prioritize to consider only the labels at the arrival event vertices of the destination depot heuristically for forbidding the branching rules, in consideration of the numerous of space–time paths. Then, in some special cases, it is necessary to check the branching rules at more vertices at previous operating stations (instead of only the destination depot), for finding alternative space–time paths that satisfy negative reduced cost and the branching rules. In this case, more columns and B&B nodes would be explored with longer computing times.

Algorithm 2 Dynamic programming for pricing sub-problems

Input:

1: The space-time network $G^R = (\mathcal{V}^R, \mathcal{A}^R)$ of rolling stock; the space-time network $G^P = (\mathcal{V}^P, \mathcal{A}^P)$ of passengers; the planning time horizon $\mathcal{T}$; current dual solutions $\pi^1 - \pi^4$ for constraints in RMP; residual cumulative passenger flow volume CP_{it}^* from the solution of RMP.

Output:

2: One best space-time path p_k^* with minimum reduced cost for rolling stock unit k .

Step 1 (Label Initialization):

4: 1. Create an origin label $RC_{d_{ori}(k), t_0} = -\pi_k^1$ to record the reduced cost at depot vertex $(d_{ori}(k), t_0)$.

5: 2. Create labels at other vertices, i.e., $RC_{it} = +\infty, (i, t) \in \mathcal{V}^R \setminus \{(d_{ori}(k), t_0)\}$.

6: 3. Create matrix C_{it} to record the traveling cost $C_{d_{ori}(k), t_0} = 0$ at depot vertex $(d_{ori}(k), t_0)$ and $C_{it} = None$ at other vertexes $(i, t) \in \mathcal{V}^R \setminus \{(d_{ori}(k), t_0)\}$.

7: 4. Create matrix PAS_{it} to record the total volume of carried passengers $PAS_{d_{ori}(k), t_0} = 0$ at depot vertex $(d_{ori}(k), t_0)$ and $PAS_{it} = None$ at other vertexes $(i, t) \in \mathcal{V}^R \setminus \{(d_{ori}(k), t_0)\}$.

8: 5. Initialize label predecessors $pred(RC_{it}) = None, i \in \mathcal{I}, t \in \mathcal{T}$.

Step 2 (Updating label dynamically):

10: **for** each timestamp $t \in \mathcal{T}$, each node $i \in \mathcal{I}$ **do**

11: **for** each successor node j of node i **do**

12: **for** traveling time TT in the travel time range of link (i, j) **do**

13: 1. Calculate the arrival time t' of node j : $t' = t + TT$.

14: 2. Choose the current label RC_{it} of space-time vertex (i, t) .

15: 3. Create a new label $RC'_{jt'}$ on space-time vertex (j, t') from label RC_{it} .

16: $RC'_{jt'} = RC_{it} + c_{i,j,t,t'} - \pi_{i,j,t,t'}^2 - \sum_{k' \in \mathcal{K} \setminus \{k\}} (\pi_{k,k',t,t'}^3 + \sum_{(t', j', t, t') \in \Omega_{i,j,t,t'}} \pi_{k,k',t',t',t,t'}^4) + Cap^k \cdot \pi_{i,j,t,t'}^4$

17: **if** $RC'_{jt'} \leq RC_{jt'}$ **then**

18: 1. Update label $RC_{jt'} \leftarrow RC'_{jt'}$.

19: 2. Update the traveling cost and total volume of carried passengers based on the inserted label.

20: $C_{jt'} = C_{it} + c_{i,j,t,t'}$

21: $PAS_{jt'} = PAS_{it} + CP_{jt'}^*$

22: 3. Update label predecessor $pred(RC_{j,t'}) = (i, t)$.

Step 3 (Backtrace):

24: 1. For rolling stock unit k , use the label predecessors $pred(RC_{it})$ and labels RC_{it} to trace back from space-time destination vertex $(d_{des}(k), t_{|T|})$ to origin vertex $(d_{ori}(k), t_0)$, and obtain a new optimal column.

25: 2. If the currently selected path (e.g., p_1) happens to be forbidden based on the branching rule (i.e., $y_{p_1} = 0$) unfortunately, then another space-time path with the second minimum reduced cost (still satisfying < 0) will be back-traced as the new column without violating the branching condition.

26: 3. Lastly, the generated new column should be pushed back to the column pool $\mathcal{P}_k$ for the subsequent iterations.

5.3. Branch-and-bound (B&B) algorithm

After solving the LP relaxation model at each branch node, if the generated solution is integral, a feasible solution to the path-based model is found; otherwise, it is needed to design an effective branching strategy to transfer the fractional variables into integral ones. To this end, some detailed branching, bounding and pruning rules will be discussed in this section to find the integral solutions.

(1) Branching rules

As for the rules for handling the fractional values of some decision variables, we firstly select a column selection variable (e.g., y_p) whose value is the nearest to 0.5 among all decision variables in the RMP model. Then, we branch y_p with two cases $y_p=0$ and $y_p=1$. That is, given the current active node n , two child nodes n_1 with $y_p=0$ and n_2 with $y_p=1$ are generated to produce the integral value for y_p . In this case, the child nodes will inherit the used columns and the branching conditions from their parent node. In addition,

in order to reduce unnecessary conflicts for subsequent iterations, all columns that conflict with space–time resources of branched and selected columns (i.e., $y_p = 1$) will be removed from the column pool. Obviously, these conflicting columns will not be selected any longer due to the safety concerns. On the other hand, columns which violate branch conditions (i.e., $y_p = 0$) will also not be generated again in the column pool with the back-tracing process based on the column checking step in the dynamic programming process (see Section 5.2.2). Lastly, the involved two newly generated branch nodes are pushed back into the active node list ANL .

(2) Bounding rules

As for the bound updating strategy in the B&B algorithm, the global lower bound (GLB) should be updated as the minimum value of the local lower bounds (LLB) on branch nodes in the active node list ANL , i.e., $GLB \leftarrow \max\{GLB, \min_{n \in ANL}\{LLB_n\}\}$. As for the global upper bound (GUB), if a feasible integer solution is found for the RMP model at a branch node, the global upper bound will be updated as the minimum of the current GUB and the newly encountered objective value in the search process.

(3) Pruning rules

In the algorithm, the following three strategies are used as pruning rules to remove the unnecessary branch nodes, aiming to speed up the searching process. (a) If we cannot find a feasible solution for the restricted master problem of the current node with pre-given branching conditions, it indicates that any possible feasible solution will not be found on this branch. For this case, we can remove this node safely from the searching tree. (b) If the optimal objective value of the current node is larger than the global upper bound, this branch will be pruned. In fact, if the optimal objective value of the current branch node is larger than the current global upper bound, the objective value of sequential branch nodes with more fixed variables will be larger than the optimal objective value of the current node. Thus, such a branch node should be removed. (c) If an optimal integer solution is founded in a branch node, the branching operation for this node will be terminated since no fractional solution can be found, and the global upper bound will be updated correspondingly.

5.4. Acceleration techniques

5.4.1. Pre-processing space–time networks of rolling stock and passengers

Obviously, the number of available rolling stock paths (i.e., columns) increases exponentially with respect to the numbers of involved stations and considered timestamps. Thus, it is better to remove some irrelevant vertexes and arcs from the space–time network so as to reduce the size of the network and simplify the subsequent optimization models. To our knowledge, time windows of pull-out/in from/to depots for rolling stock can be well specified in constructing the involved space–time network for the real-world operations. For example, most rolling stock units need to pull out from their depots in the morning (e.g., 6:00–8:00) and pull in to their home depots at night (e.g., 21:00–23:00). At the same time, a small number of rolling stock units are scheduled to pull out/in from/to depots in off-peak hours in order to satisfy the peak-hour passenger demand (e.g., 6:00–10:00, 17:00–22:00). To this end, some space–time pull-out/in arcs outside the involved time windows could be removed from the network to avoid generating columns with undesirable operation rules. Moreover, considering the homogeneous characteristics for all rolling stock units with the same type, the pull-out/in operations of rolling stock from/to depots can be scheduled in a sequential order. Accordingly, the starting time of pull-out/in time windows for each rolling stock unit can be specified with the designed intervals.

As for the space–time network for passenger flow movement, the volumes of inflow passengers at vertexes should be balanced in order to make flow balance constraints (5)–(8) feasible. However, in some small and medium scale cases, when the planning time horizon is set as a short period (e.g., 7:00–10:00), the total number of inflow passengers at involved stations possibly does not equal the total number of outflow passengers, leading to some incomplete trips of passengers during the considered time horizon (Mo et al., 2021). In order to ensure all passengers can get on one rolling stock unit and reach their destinations without violating passenger flow balance constraints, some vertexes and arcs related to incoming passengers in the period, in which some passengers cannot arrive at their stations with complete trips during the given time horizon, need to be removed from the network in order to keep the balance of inflow and outflow passengers.

5.4.2. Parallel computing techniques for B&B process

With the branching, bounding, and pruning rules, the branch-and-bound algorithm can be implemented for searching high-quality solutions for the proposed model. Compared to the common sequential computing in the B&B procedure, **two** parallel computing techniques are considered for the B&B tree in this study to improve the solution efficiency for the involved model by collaboratively utilizing the multiple-core computation resources.

(1) Parallel BFS and DFS techniques (Outer parallel computation)

Generally, different B&B searching strategies can also affect the algorithm performance. To our knowledge, the depth-first-search (DFS) strategy is helpful to improve the global upper bound by quickly finding feasible integral solutions, while the breath-first-search (BFS) strategy is helpful to improve the global lower bound by exploring more extensive branch nodes (Lu et al., 2022). Thus, in this study, a B&B algorithm with parallel computing techniques is presented with both BFS and DFS searching strategies (i.e., outer parallel computation). Specifically, the BFS and DFS implementation codes are processed in parallel. At the same time, these two processes will exchange status information in upper and lower bound updating steps. That is, when a feasible solution with an effective UB is obtained in the DFS process, the corresponding UB in the BFS process will also be updated in real time, which can prune more branch nodes by using such UB from the DFS thread. Similarly, the LB in DFS will also be updated in correspondence with the obtained LB from the BFS process in real time, which can help to improve the optimality of the obtained feasible solution in the DFS process.

Algorithm 3 Parallel branch-and-bound algorithm

```

1: function MAIN_FUNCTION
2:   1. Initialize global upper bound  $GUB = +\infty$  and global lower bound  $GLB = -\infty$ .
3:   2. Create one origin node ( $origin\_node$ ).
4:   # parallel for (Outer parallel computation)
5:   —BREADTH_FIRST_SEARCH (BFS)( $origin\_node$ )
6:   —DEPTH_FIRST_SEARCH (DFS)( $origin\_node$ )
7:
8: function BREADTH_FIRST_SEARCH (BFS)( $origin\_node$ )
9:   Push back  $origin\_node$  into the active node list  $ANL$ :  $ANL \leftarrow \text{PUSH}(ANL, origin\_node)$ .
10:  while True do
11:    if  $ANL == \emptyset$  or  $gap = (GUB - GLB)/GUB < \epsilon$  then
12:      Break
13:    1. Initialize local upper bound  $LUB = GUB$  and local lower bound  $LLB = GLB$  on the current layer.
14:    2. Distribute all nodes in  $ANL$  to  $m$  processors.
15:    # parallel for  $m$  processors (Inner parallel computation)
16:    —Solve nodes parallelly and averagely in each processor  $m$ , and obtain upper bound / lower bound / child nodes on each node.
17:    3. Update local upper bound  $LUB$  / local lower bound  $LLB$  with the minimum lower bound / upper bound within all solutions of nodes.
18:    if  $LUB < GUB$  then
19:       $GUB = LUB$ 
20:    if  $LLB > GLB$  then
21:       $GLB = LLB$ 
22:    4. Push back all child nodes from different processors into  $ANL$ .
23:
24: function DEPTH_FIRST_SEARCH (DFS)( $origin\_node$ )
25:   1. Initialize local upper bound  $LUB = GUB$  and local lower bound  $LLB = GLB$ .
26:   2. Initialize the active node stack  $ANS = \emptyset$ .
27:   3.  $cur\_node \leftarrow origin\_node$ 
28:   while True do
29:     if  $ANS == \emptyset$  or  $gap = (GUB - GLB)/GUB < \epsilon$  then
30:       Break
31:     if  $cur\_node \neq None$ : then
32:       Solve the  $cur\_node$  and obtain objective value  $obj$  and solution  $y^*$ .
33:       if  $obj < GUB$  then
34:         if  $y^*$  is an integer solution then
35:            $GUB = LUB$ 
36:            $cur\_node \leftarrow None$ 
37:         else
38:           1. Create a left child nodes  $left\_node$  from  $cur\_node$  based on the branching rules.
39:           2.  $ANS \leftarrow \text{PUSH}(ANS, cur\_node)$ 
40:           3.  $cur\_node \leftarrow left\_node$ 
41:       else
42:         1.  $cur\_node \leftarrow \text{POP}(ANS)$ 
43:         2. Create a right child nodes  $right\_node$  from  $cur\_node$  based on the branching rules.
44:         3.  $cur\_node \leftarrow right\_node$ 

```

(2) Parallel technique for same layer nodes (Inner parallel computation)

Furthermore, in the BFS implementation, the nodes on the same branching layer with the BFS strategy will be further solved in a parallel manner independently (i.e., inner parallel computation). Specifically, when all nodes are obtained on one branching layer, these nodes will be distributed equally into multiple processors based on the number of CPU threads. After solving those nodes in different processors, we can observe that the BFS lower bound can be updated based on the minimum objective of nodes on the same layer. Thus, no matter in the serial or parallel BFS process, the lower and upper bounds are updated after solving all nodes on one branch layer. Therefore, they can obtain the same solutions and final results with the same node enumerations, and the only difference lies in the computation sequence of independent nodes on one layer.

A brief introduction of the branch-and-bound with parallel computing is presented in Algorithm 3. Note that hashtag “# parallel for” indicates the number of parallel threads that are created for each decomposed task, and threads have to wait here for all threads to end at the joint point. Then, tasks with the operator “—” can be processed simultaneously by multiple threads. For clarity, an illustrative instance is shown in Fig. 11 to better interpret such a parallel process with a partial three-layer B&B tree on the last one and last but one layer. Specifically, BFS upper bounds are updated additionally after solving nodes $k + 2, k + 3, k + 4, k + 5$ in the DFS process, and DFS lower bounds are updated additionally after solving node layers 1 and 2 in the BFS process. Obviously, in comparison with the serial B&B process, a faster upper and lower bound updating process is implemented in the parallel B&B algorithm.

6. Numerical experiments

In this section, a series of numerical experiments, involving an illustrative small-scale case study and a real-world case study on the Batong line of the Beijing metro system, are conducted to testify the effectiveness of proposed models and algorithms. All

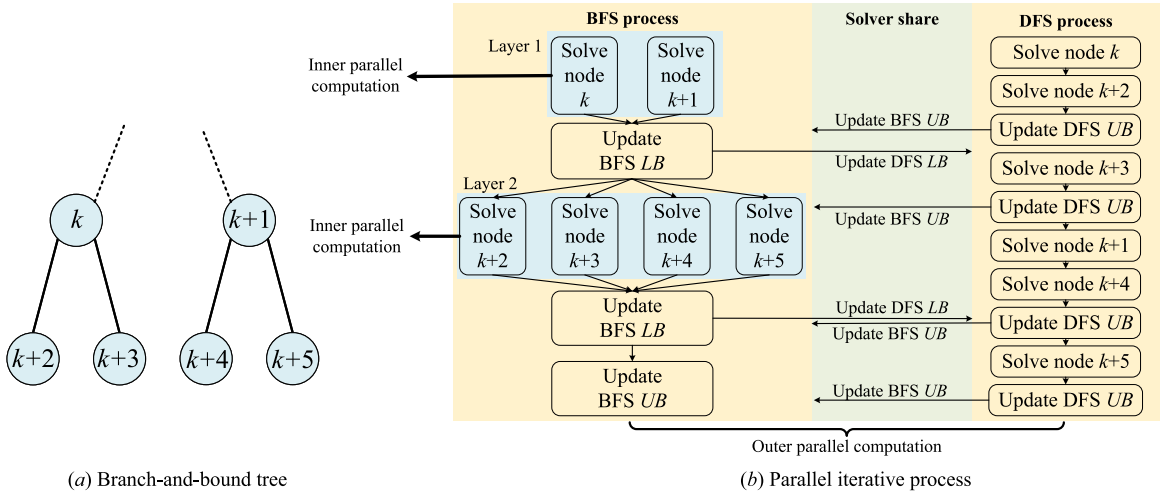


Fig. 11. Iterative process of the parallel B&B algorithm.

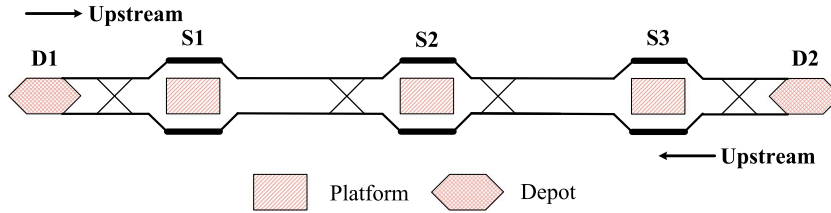


Fig. 12. Bi-directional metro line in the small-scale case study.

numerical experiments are implemented in Python 3.8.12 and Gurobi 9.5.0 on a computer with Intel(R) Core(R) i7-10700@ 2.60 GHz CPU and 16 GB RAM. The small-scale cases are solved by both the proposed approach and Gurobi solver for the convenience of comparison, while the large-scale cases are tested by our proposed B&P algorithm due to the difficulty of solving the arc-based model by solvers.

6.1. Small-scale case study

In this part, we consider a bi-directional metro line with three stations in each direction, shown in Fig. 12. For the convenience of description, we particularly name the stations as S1, S2 and S3 respectively, i.e., $I = \{S1, S2, S3\}$, in which the upstream direction is from stations S1 to S3, and the downstream direction is from stations S3 to S1. Besides, two depots D1 and D2 are built to connect with stations S1 and S3, respectively. A total of four rolling stock units, i.e., three in depot D1 and one in depot D2 initially, are taken into account in the experiments. All stations S1, S2 and S3 are assumed as operating stations, where rolling stock can be coupled or decoupled on turn-around tracks. All rolling stock units are required to return back to their home depots at the end of the planning time horizon.

To our knowledge, smaller time granularities can provide more accurate solutions but consume longer computing time, and just the opposite with larger time granularities. Based on some preliminary experiments of different time discretization granularities in Appendix, we find that the benefits of computing time-saving get much higher than the benefits of objective values for two cases with 45 s and 60 s granularities. However, when the time granularity is further increased to the 90 s, the benefit in computing time only decreases by about 10% with about a 3% increase in the objective function. Therefore, similar to the studies (e.g., Pan et al., 2023), the discretization granularity with 1 min (60 s) is selected in numerical experiments for balancing the computing time and the solution accuracy. Thus, before solving the proposed mathematical models, we first denote the discretized time horizon as $T = \{1, 2, \dots, 40\}$ (unit: min). Then, some input parameters with respect to the metro line and rolling stock units are set to construct the space-time network, as summarized in Tables 4 and 5. Particularly, the time windows of pull-in/out operations for rolling stock are specified in order to reduce the scale of the space-time network. The operation cost for each rolling stock unit is set to \$35/min and attached to the cost of each space-time arc, i.e., $c_{i,j,t,t'} = \$35 \cdot (t' - t)$. Lastly, the maximum number of carried passengers for one rolling stock unit is assumed to be 380 (unit: person/rolling stock unit). Based on the constructed space-time network for rolling stock, the proposed arc-based and path-based models can be formulated, where the weight coefficients α and β in the objective function (i.e., Eq. (26)) are all set as 1.

Table 4
Parameters for space–time network construction.

Parameter	Value
Dwell time range at stations S1, S2 and S3	[1 min, 2 min]
Running time range on segments S1–S2, S2–S3	[3 min, 4 min]
(De)coupling time range at stations S1, S2, and S3	[1 min, 2 min]
Turn-around time at stations S1, S2, and S3	4 min, 3 min, 4 min
Minimum headway time	3 min
Pull-in/out time to/from depots for rolling stock	4 min

Table 5
Parameters for the operations of rolling stock.

Rolling stock index	Home depot	Pull-out time window	Pull-in time window	Operation cost	Capacity (person / rolling stock unit)
1	D1	[0 min, 5 min]	[0 min, 40 min]	\$35/min	380
2	D1	[5 min, 10 min]	[0 min, 40 min]	\$35/min	380
3	D1	[0 min, 10 min]	[0 min, 40 min]	\$35/min	380
4	D2	[0 min, 10 min]	[0 min, 40 min]	\$35/min	380

Regarding the passenger demand, to create an over-saturated scenario and investigate performances over different passenger demand scenarios, we consider the baseline passenger demand and the additional passenger demand as the experiment data. We here assume that the inflow passengers only appear from timestamp 2 to timestamp 16. Then, the baseline passenger demand can be represented by two 3×16 volume matrixes for inflow passengers in the upstream direction (i.e., $\mathbf{P}^{\text{in(up)}}$) and the downstream direction (i.e., $\mathbf{P}^{\text{in(down)}}$) given below. In addition, the total numbers of outflow passengers at stations S1, S2 and S3 are assumed to be 0, 900 and 450 in upstream direction, respectively, while these numbers in downstream direction are 60, 60 and 0. Moreover, to generate a congestion scenario for the problem of interest, a total of 540 extra passengers at station S1 on timestamp 2 with OD S1→S2 are considered as the additional passenger demand, i.e., the number of inflow passengers at station S1 on timestamp 2 is $60+540=600$ in the experiment.

$$\mathbf{P}^{\text{in(up)}} = \left[\mathbf{P}_{it}^{\text{in(up)}} \right]_{3 \times 16} = \begin{pmatrix} 0 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 & 60 \\ 0 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 & 30 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} S1 \\ S2 \\ S3 \end{matrix}$$

$$\mathbf{P}^{\text{in(down)}} = \left[\mathbf{P}_{it}^{\text{in(down)}} \right]_{3 \times 16} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 30 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 30 & 0 & 0 & 0 & 0 \\ 0 & 30 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 30 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} S1 \\ S2 \\ S3 \end{matrix}$$

6.1.1. Computational results

In the experiment, we firstly formulate the arc-based model for the problem of interest. Note that the arc-based model is a mixed integer linear programming model, we then solve it by using Gurobi solver, in which the termination criteria are set as follows: (a) when a solution within the optimality gap 2% is found, the calculation will be terminated; (b) the searching process can be ended when the computing time is over 20 s. After implementations, we find a near optimal solution with the computing time 17.83s, where the optimal value turns out to be 4865.58 with gap=0.00%.

To improve the computational efficiency, we next use the proposed B&P algorithm to solve the path-based model with the parallel computing technique. In the solution process, the first step is to generate an initial column pool by using the DP algorithm. After implementing the proposed B&P algorithm, four columns are selected to form a feasible solution with the objective value of 4865.57, which is the same with that of the arc-based model, and this optimal feasible solution is obtained with a computing time of 9.75s and a relative gap of 1.85%. In addition, the visualization of this solution and the corresponding passenger flow distribution are shown in Fig. 13, in which the lines with different colors denote the detailed space–time paths (columns) for different rolling stock units. In particular, all the shadows filled with red colors denote the volume of cumulative passengers on station platforms in the upstream direction, and those filled with blue colors denote the volume of cumulative passengers in the downstream direction. The double lines denote the trips with large train compositions (i.e., two coupled rolling stock units). We observe that the operation scheme in this result is consistent with passenger demand. To be specific, rolling stock units 1 and 2 are only scheduled to service segment S1–S2 instead of the whole metro line, since much larger passenger demand is distributed on segment S1–S2 than segment S2–S3. Also, rolling stock units 1 and 3 are coupled in depot D1 on timestamp 1 and decoupled at station S2 on timestamp 10 in order to satisfy the over-saturated demand at station S1 during the early time period.

To further investigate the performance of our methods, we implement this experiment particularly by using serial computation. The comparison for the computational results by the parallel computing technique, serial computing technique and Gurobi solver can be referred to Table 6. When the same optimal termination condition is met, the computing time by using the parallel computing technique (i.e., 7.84s) can be reduced by 60% than that of the path-based model solved by using the serial computing technique (i.e., 19.74s), and can be reduced by 54% than that of the arc-based model solved by Gurobi directly (i.e., 17.16s). In fact, in comparison to the solution process with the serial computing technique, multiple nodes in the same branching layer can be solved simultaneously in parallel computing with multiple CPU processors, resulting in a significant reduction of the computing time.

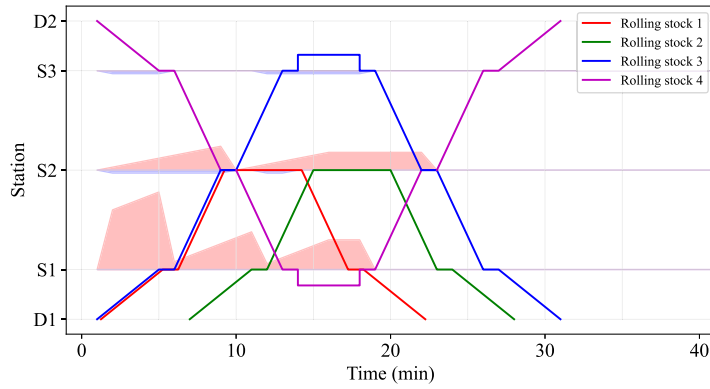


Fig. 13. Visualization of the near optimal solution in the small-scale case.

Table 6

Computational results with different solution methods for the small-scale case.

Method	Computing technique	Objective value			Computing time (s)
		Lower bound (LB)	Upper bound (UB)	Gap (%)	
B&P	Serial	4775.49	4865.57	1.85	19.99
Gurobi	—	4865.57	4865.57	0.00	17.83
B&P	Parallel	4775.49	4865.57	1.85	9.75

Table 7

Computational results for different instances.

Instance index	Instance feature abbreviation	Instance feature			B&P					Gurobi solver				Gap 3 (%)
		Short-turning strategy	Train composition	Additional passenger demand	Lower bound (LB_1)	Upper bound (UB_1)	Gap 1 (%)	Computing time (s)		Lower bound (LB_2)	Upper bound (UB_2)	Gap 2 (%)	Computing time (s)	
								Parallel computation	Serial computation					
1	NST_C1_P1	NST	C1	P1	5468.31	5478.42	0.18	8.37	12.68	5478.42	5478.42	0.00	16.71	0.00
2	NST_C1_P2	NST	C1	P2	5063.72	5282.50	4.14	9.25	41.94	5076.59	5076.59	0.00	15.28	3.90
3	NST_C2_P1	NST	C2	P1	5354.01	5354.01	0.00	2.47	3.35	5354.01	5354.01	0.00	17.67	0.00
4	NST_C2_P2	NST	C2	P2	5010.71	5199.56	3.63	9.69	15.57	4968.20	5030.83	1.24	14.71	3.25
5	ST_C1_P1	ST	C1	P1	5024.30	5093.62	1.36	6.67	10.36	4941.26	5028.60	1.74	15.05	1.28
6	ST_C1_P2	ST	C1	P2	4564.70	4719.71	3.28	6.46	6.89	4472.30	4603.89	2.86	20.00	2.45
7	ST_C2_P1	ST	C2	P1	4775.49	4865.57	1.85	9.75	19.99	4865.58	4865.58	0.00	17.83	0.00
8	ST_C2_P2	ST	C2	P2	4441.18	4603.88	3.53	9.99	47.12	4340.79	4533.82	4.26	20.00	1.52
						Ave.	2.25	7.84	19.74		Ave.	1.26	17.16	1.55

Notes for Table 7:

(1) Short-turning strategy: Allowed (ST); not allowed (NST).

(2) Train composition: Maximum 1 rolling stock unit in a train composition (C1); Maximum 2 rolling stock units in a train composition (C2).

(3) Additional passenger demand: Scenario 1 (P1); scenario 2 (P2).

(4) Gap 1: Relative gap between UB_1 and LB_1 in the proposed B&P algorithm.(5) Gap 2: Relative gap between UB_2 and LB_2 in Gurobi.(6) Gap 3: The gap between UB_1 of B&P and UB_2 of Gurobi.

6.1.2. Performance analyses with different instances

We note that the computational results of our proposed approaches are closely related to the specific operation schemes. E.g., if different operation patterns and passenger distributions are considered in the problem, the computational results will also be different. It is thus worth investigating the performance of our proposed approaches with a variety of rolling stock operation patterns and passenger demand scenarios. To this end, we next conduct a total of eight extra experiments with different parameter information. For convenience, we introduce an index with the form “A_B_C” to denote the experiment characteristics, in which “A” represents whether the short-turning action is allowed at station S2 (i.e., ST: allowed, NST: not allowed), “B” represents the maximum number of rolling stock units in a train composition (i.e., C1: one rolling stock unit, C2: two rolling stock units), and “C” represents the additional passenger demand scenario (i.e., P1: 540 additional passengers at station S1 on timestamp 2 with destination S2, P2: 540 additional passengers at station S1 on timestamp 11 with destination S2). For example, instance “ST_C1_P1” shows the experiment settings as follows: considering the short-turning action at station S2, setting the maximum number of rolling stock units in a train composition as 1, and considering additional passenger demand at station S1 on timestamp 2.

For the convenience of comparison, we use both the Gurobi solver and our B&P algorithm with the parallel and serial computing techniques to solve these instances. Table 7 lists the computational results for different tested instances, including the optimal objective values (lower and upper bounds), gaps and computing time. In these experiments, the proposed B&P method with parallel computing can find near optimal solutions using much less time than the Gurobi solver for all instances (see column “Computing time (s)” for details). Next, we will elaborate the performance analyses with respect to different model settings.

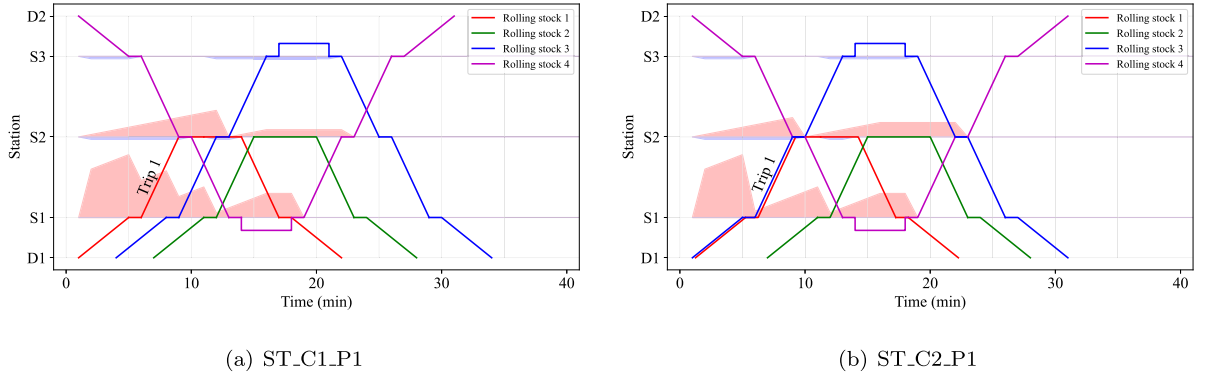


Fig. 14. Comparison of computational results with different operation modes.

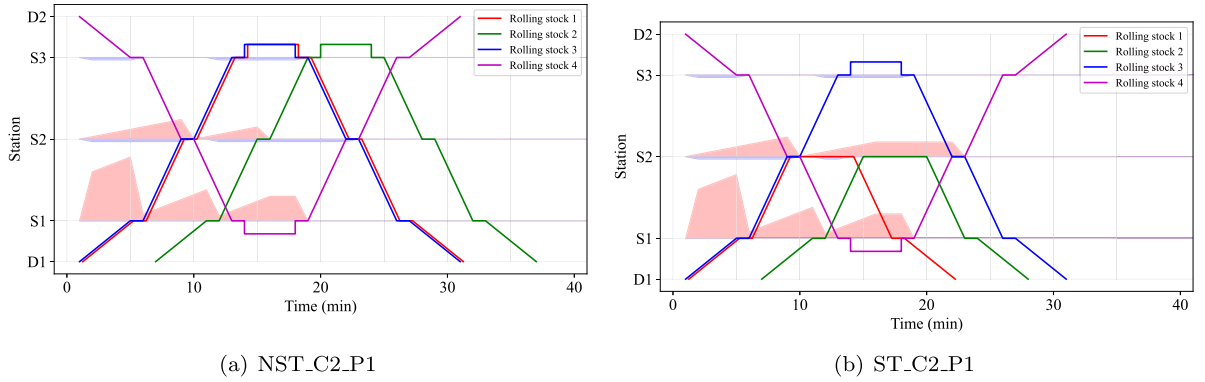


Fig. 15. Comparison of computational results with different short-turning operation strategies.

(1) Performance analyses with respect to the maximum number of rolling stock units in a train composition. Compared with the fixed composition mode, the flexible composition mode with multiple coupled rolling stock units might correspond to a better objective, which also improves the passenger service with less passenger waiting costs. For example, when additional passenger demand scenario P1 is considered, a large train composition with two coupled rolling stock units is scheduled for trip 1 in instance “ST_C2_P1” in Fig. 14(b). This solution corresponds to a lower total objective value (i.e., 4865.57) with less congestion at station S1. Conversely, in case “ST_C1_P1” without the flexible operation mode in Fig. 14(a), some passengers cannot get on trip 1 due to the capacity limit of the small composition train, and need to wait for subsequent arriving trains, leading to higher total objective value (i.e., 5093.62) and lower passenger service level. Thus, when the passenger demand is uneven in spatial and temporal dimensions, it is a beneficial way to match the passenger distribution by flexibly assigning rolling stock for different trips, so as to improve service levels and maximize the utilization of rolling stock sufficiently.

(2) Performance analyses with respect to short-turning operation strategies. The short-turning strategy has a crucial impact on the space–time paths of rolling stock. When the large passenger demand occurs on some segments, more rolling stock units can be scheduled with less headway or larger composition on these busy areas instead of the whole metro line, which might reduce the operation costs and match the passenger distribution more precisely. For example, a significant difference in the schedules can be deduced between instances “NST_C2_P1” in Fig. 15(a) and “ST_C2_P1” in Fig. 15(b). In instance “NST_C2_P1”, when a total of 540 additional passengers are taken into consideration at station S1 on timestamp 2, segment S1→S2 in the early operation stage is typically in an over-saturated state. Since the short-turning operation is not allowed at station S2, all rolling stock units have to turn around at station S3 and pass through segment S2→S3 with less onboard passengers in Fig. 15(a), resulting in a waste of transport resources on this segment and a larger objective value 5354.01. In comparison, when the short-turning operation is permitted at station S2 in case “ST_C2_P1”, rolling stock units 1 and 2 can turn around at station S2 and then service segment S2→S1 in Fig. 15(b), which corresponds to a less objective value 4865.57 than that of instance “NST_C2_P1”. Likewise, similar results are also obtained in other instance pairs, e.g., instances 1 vs. 5. The above computational results demonstrate that the short-turning operation strategy can correspond to better objective values and improve the circulation efficiency for rolling stock units with the same experimental conditions. Thus, to improve the operation efficiency with less operation costs, crossover tracks are encouraged to be built at some appropriate stations to facilitate the flexible operations of trains, with the condition of satisfying all the passenger demand.

(3) Performance analyses with respect to additional passenger demand scenarios. In the real-world applications, the rolling stock schedule should be closely related to the distribution of time-varying passenger demand. Therefore, it is necessary

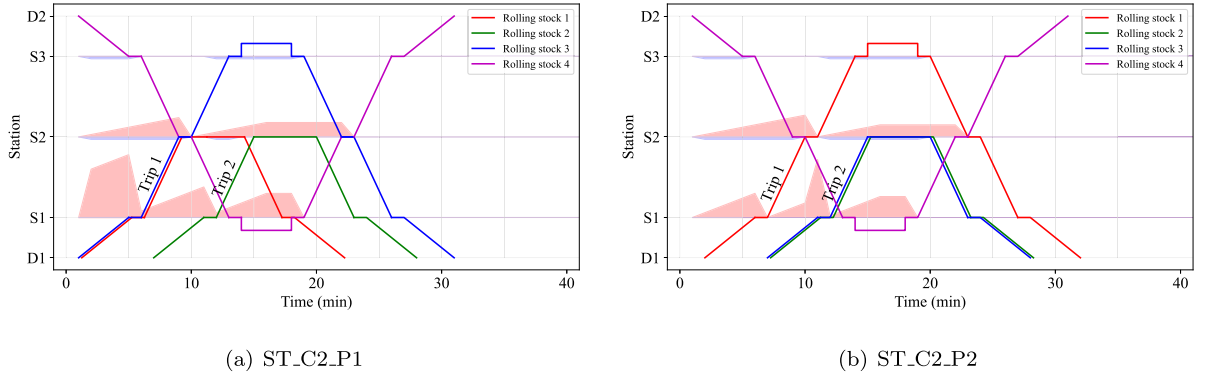


Fig. 16. Comparison of computational results with different passenger demand scenarios.

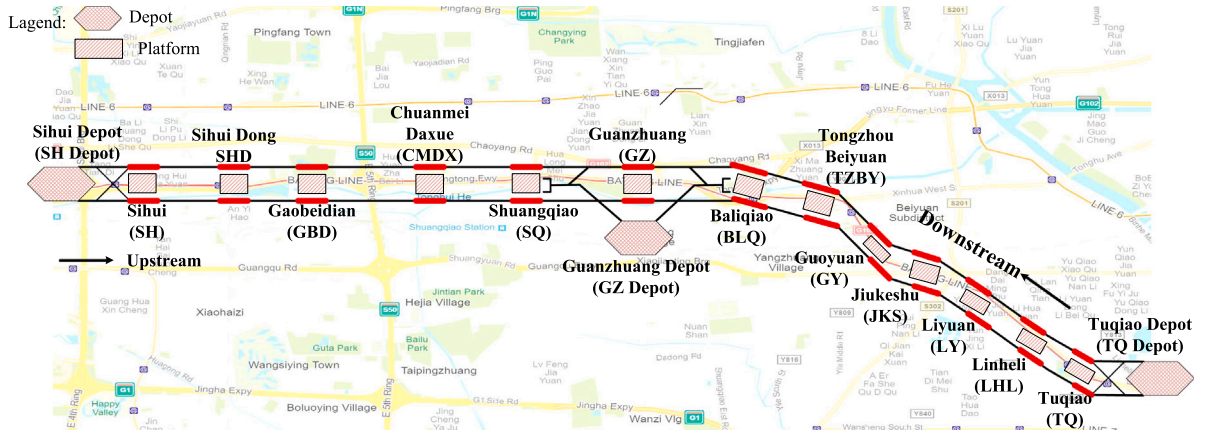


Fig. 17. Overview of Beijing metro Batong line.

and interesting to investigate the performances of the proposed approaches under different passenger demand scenarios. With the computational results in this set of experiments, we observe that the large train composition is in general scheduled to service space–time vertices with larger passenger demand. For example, in case “ST_C2_P1”, the large train composition is assigned to trip 1 on timestamp 1 in Fig. 16(a) due to the arrivals of additional passengers on timestamp 2; while the large train composition is assigned to trip 2 on timestamp 8 in Fig. 16(b) due to the occurrence of additional passengers on timestamp 11 in case “ST_C2_P2”. In addition, the objective value of case “ST_C2_P2” (i.e., 4603.88) is less than that of case “ST_C2_P1” (i.e., 4865.57), which also implies that additional passengers occurring in later time periods have smaller impacts for the total cumulative passengers and waiting costs. According to these analyses, the flexible train composition schemes could be well optimized according to different passenger demand scenarios, further validating the effectiveness of the proposed model and algorithm.

6.2. A large-scale case study

To further demonstrate the performance of the proposed approach for large-scale cases, we next consider a case study based on the Beijing metro Batong line, which is a bi-directional metro line with 13 stations. Fig. 17 shows the detailed track layouts and segment lengths, where the downstream direction is from “Tuqiao” to “Sihui” stations and the upstream direction is from “Sihui” to “Tuqiao” stations. We assume that there are three depots along this line, which connect with three operating stations, i.e., Sihui, Guanzhuang and Tuqiao, at which the turn-around, coupling and decoupling activities are permitted. For constructing the space–time network in convenience, some parameters of this metro line is displayed in Table 8.

Regarding the passenger demand, the time-varying passenger flow data are all taken from historical operation data in the Auto Fare Collection (AFC) system. Specifically, the inbound passenger flow P_i^{in} can be recorded directly at each station i on each timestamp t , and the total outbound flow P_i^{out} can be obtained by summing all outbound passengers along the planning time horizon at station i . Fig. 18 displays the time-varying inbound volume of passengers at each station on the Beijing metro Batong line. As shown, the number of passengers in the downstream direction (i.e., from Tuqiao to Sihui) is significantly larger than that in the upstream direction (i.e., from Sihui to Tuqiao) in the morning peak hours and the opposite characteristics occur in the evening peak hours. At the same time, the passenger demand at each station also presents two “uni-modal” distributions,

Table 8
Parameters for space–time network construction.

Segment / Station	Operation event	Operation time (min)	Segment / Station	Operation event	Operation time (min)	Segment / Station	Operation event	Operation time (min)
Depot-SH	Pull-in/out	4	SQ-GZ	Running	[2,3]	JKS	Dwell	1
SH	Turn around	[4,6]	GZ	Turn around	[4,6]	JKS-LY	Running	[2,3]
	Dwell	1		Dwell	1	LY	Dwell	1
SH-SHD	Running	[2,3]	Depot-GZ	Pull-in/out	4	LY-LHL	Running	[2,3]
SHD	Dwell	1	GZ-BLQ	Running	[2,3]	LHL	Dwell	1
SHD-GBD	Running	[2,3]	BLQ	Dwell	1	LHL-TQ	Running	[1,2]
GBD	Dwell	1	BLQ-TZBY	Running	[2,3]	TQ	Turn around	[4,6]
GBD-CMDX	Running	[3,4]	TZBY	Dwell	1		Dwell	1
CMDX	Dwell	1	TZBY-GY	Running	[2,3]	TQ-Depot	Pull-in/out	4
CMDX-SQ	Running	[2,3]	GY	Dwell	1			
SQ	Dwell	1	GY-JKS	Running	[1,2]			

Note for Table 8:

(1) Segment/Station: For simplicity, abbreviations for station names are used in this table, e.g., Sihui station is shortened to SH.

(2) Operation time: The operation time range is denoted by the form “[a, b]”, in which “a” and “b” represent the minimum and maximum operation time durations.

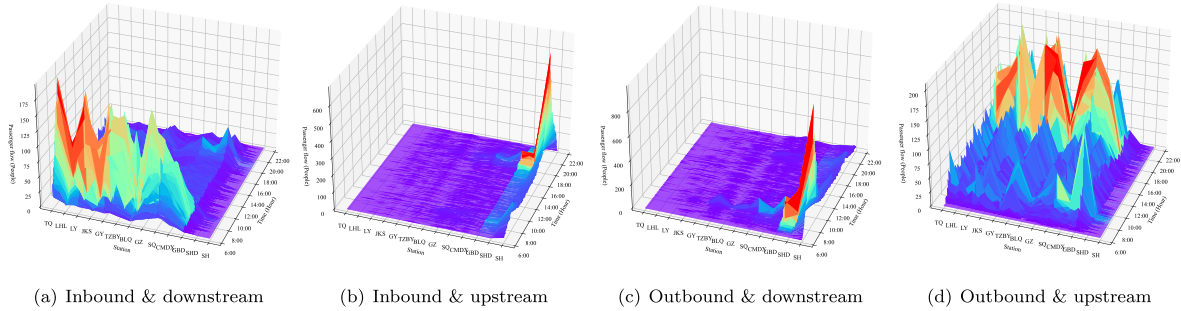


Fig. 18. Illustration of time-varying passenger flow on Beijing metro Batong line.

Table 9
Parameters for operations of rolling stock.

Rolling stock operation	Depot	Time window	Rolling stock operation	Depot	Time window
Morning regular pull-out	Tuqiao/Sihui	[6:00, 7:30]	Evening peak-hour pull-out	Sihui	[17:00, 18:30]
Morning peak-hour pull-out	Tuqiao	[6:00, 7:30]	Evening peak-hour pull-in	Tuqiao	[20:30, 22:00]
Morning peak-hour pull-in	Sihui	[8:30, 10:00]	Evening regular pull-in	Sihui/Tuqiao	[21:30, 23:00]

where most passenger demand concentrates on the periods during 6:00–10:00 in the downstream direction and 17:00–22:00 in the upstream direction. Therefore, the flexible train composition mode could easily adapt to such a passenger flow distribution pattern by allocating long compositions during morning/evening peak periods and short compositions during noontime off-peak periods, with service improvements and operation cost savings.

6.2.1. Computational results

To begin with, we firstly give the detailed experiment descriptions and parameter settings. In this set of experiments, the planning time horizon is considered as time interval [6:00, 23:00], which covers morning/evening and noontime off-peak periods. To construct the space–time network for rolling stock and passengers, the time horizon is particularly discretized into a total of 510 timestamps by a time granularity of 2 min. Particularly, in this case, only Tuqiao and Sihui stations are taken into consideration as operating stations with depots, and the experiments with more depots can refer to Section 6.2.2. Besides, a total of 20 rolling stock units are stored in Sihui and Tuqiao depots. Then, time windows and related depots of pull-out/in operations for rolling stock are specified in order to reduce the scale of the space–time network, as displayed in Table 9. Moreover, the operation cost for each rolling stock unit is set to \$35/min (i.e., $c_{i,j,t,t'} = \$35 \cdot (t' - t)$). Lastly, the maximum number of carried passengers in one rolling stock unit is assumed to be 2920 persons with B-type units.

In the experiment implementations, we firstly use Gurobi solver to solve the proposed arc-based model formulated in the corresponding space–time network. Nevertheless, when the computing time reaches 7200s, Gurobi solver cannot find even a feasible solution due to the enormous numbers of decision variables (i.e., 0–1 space–time arc selection variables) and constraints. With this concern, we turn to the developed B&P algorithm to solve the proposed path-based model in this set of experiments.

To improve the computational efficiency in the solution process, three acceleration techniques are embedded into the B&P algorithm in order to generate high-quality solutions with an acceptable computing time: (1) To avoid the long tail effect in the lower bound updating process, the column generation process is terminated when the reduction of objective values is less than 5% in the last 5 iterations. (2) A pre-defined number of branch nodes on one layer with smallest objectives are reserved in subsequent

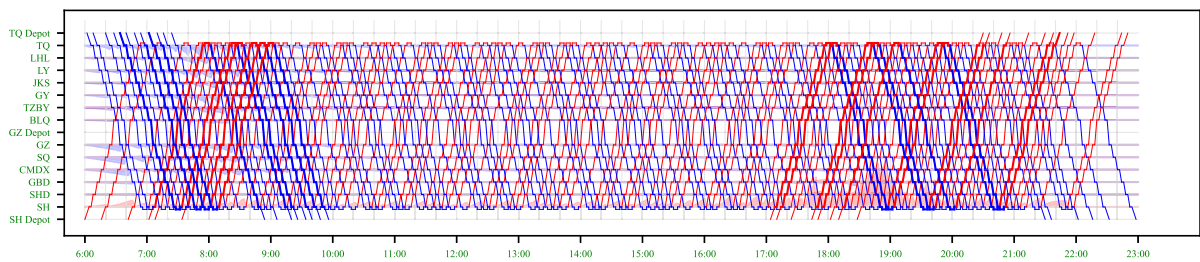


Fig. 19. Visualization of generated rolling stock schedule on Beijing metro Batong line.

Table 10

Summary of optimization results in different instances.

Instance index	Scenario information							Computational performance						
	# of depots	# of rolling stock units storage of depots			Operating station	Rolling stock type	Flexible composition mode	# of trips	# of B&B nodes	Secondary waiting passenger (person-min)	Obj.val	Computing time (s)		
		SH	GZ	TQ								Parallel computation (PC)	Serial computation (SC)	Reduction (%)
1	2	14	0	6	SH, TQ	B-type	Y	223	503	18 499	731 004.40	5573.62	26 999.29	79.36
2	1	20	0	0	SH	B-type	Y	205	341	27 372	974 200.58	3378.38	17 439.11	80.63
3	3	8	6	6	SH, TQ, GZ	B-type	Y	260	371	3173	675 594.10	4188.52	20 817.97	79.88
4	2	14	0	6	SH, TQ	B-type	N	226	1800	4152	1120144.56	7200.00	28 800.00	75.00
5	2	14	0	6	SH, TQ	A-type	Y	227	1646	6747	711 501.86	4228.21	22 960.75	81.59
Ave.										842 489.10	4913.75	23 403.42	79.29	

Notes for Table 10:

(1) # of depots: 1 depot attached to station SH; 2 depots attached to stations SH and TQ; 3 depots attached to stations SH, TQ and GZ.

(2) # of rolling stock units storage of depots: A total of 20 rolling stock units are allocated for all instances.

(3) Rolling stock type: A-type rolling stock with a capacity of 3440 passengers (A-type); B-type rolling stock with a capacity of 2920 passengers (B-type).

(4) Flexible composition mode: In order to provide adequate supply services for passenger demand, all trips will be covered by long train compositions in the fixed composition mode for instance 4.

(5) Computing time reduction = $\frac{SC-PC}{SC} \times 100\%$.

(6) The solution algorithm process is terminated when the computing time is over 7200 s in parallel computation and 28800 s in serial computation.

branch-and-bound iterations, for keeping a relative small number of nodes on one layer and finding better feasible solutions quickly. Specifically, the number of branch nodes on one layer can be estimated approximately based on the acceptable computing time. At the same time, for the convenience of distributing nodes in parallel computation and back-tracing their parent nodes with a balanced binary tree, the number of branch nodes is better to be set with some power of two (i.e., 8, 16, 32, 64, ...) in our study. (3) To avoid too many branch nodes on rarely used columns in later branching iterations, the column whose selection variable (i.e., y_p) takes 0 in the RMP model will be removed from the column pool.

When the relative gap of the branch-and-bound process with the acceleration techniques reaches a threshold of 5%, we find a near optimal solution for this instance with the objective value 731004.40. For clarity, Fig. 19 displays the optimized rolling stock scheduling scheme and the corresponding passenger accumulation situation at each station, where a trip with overlapping bold lines indicates that this trip will be serviced by a large composition with two connected rolling stock units. In this solution, it is easy to see that similar and smooth congestion conditions appear during the peak and off-peak periods by arranging long rolling stock compositions along morning and evening peak periods. Specifically, during the morning peak period 6:00–10:00, a total of 16 trips are operated with two connected rolling stock units to satisfy the over-saturated passenger demand. In comparison, during the off-peak period 10:00–17:00, the large train composition is not indispensable generally because the passenger demand can be well satisfied by setting short train compositions. At the same time, some rolling stock units are stored in the depot attached to the Sihui station during the noontime off-peak demand period. Lastly, these rolling stock units depart from Sihui depot and end at Tuqiao depot with 19 long composition trips for keeping the rolling stock conservation at the beginning and end of one day. In comparison with the fixed composition mode (i.e., instance 4 in Table 10), the total objective value is significantly reduced by 34.74% with the flexible train composition mode, since numerous short compositions are scheduled during the off-peak hours for rolling stock cost savings.

Similar to the small-scale case studies, to further investigate the performance of our proposed methods, we also implement the same experiment by using the serial computing technique for comparison. When the same termination condition is met, the computing time by using the serial computing technique (i.e., 26999.29s) is enlarged by 79.36% than that by using the parallel computing technique (i.e., 5573.62s), since the generated nodes can only be solved sequentially with one CPU processor.

6.2.2. Performance analyses with different experiment instances

With the computational and visualization results, some new characteristics are evident in the large-scale case study in addition to those in small-scale cases. It is thus worthwhile to investigate the performance of our proposed approaches from a more macroscopic level in large-scale cases. To this end, we next consider a total of six extra experiment instances with different parameter information, including (a) different numbers of depots, (b) rolling stock capacities, and (c) passenger demand distribution scenarios.

Table 11
Summary of optimization results in different instances.

Instance index	Scenario information						Computational performance				
	# of depot	# of rolling stock units storage of depots			Passenger demand scenario	Planning time horizon	Secondary waiting passenger (person-mia)	Obj. val	Computing time (s)		
		SH	GZ	TQ					Parallel computation (PC)	Serial computation (SC)	Reduction (%)
6	2	12	0	8	P2	[6:45–10:15]	2243	155 725.36	7200.00	28 800.00	75.00
7	2	12	0	8	P3	[11:45–15:15]	1302	82 407.06	5263.50	24 856.23	78.82
Ave.								119 066.21	6231.75	26 828.12	76.91

Notes for Table 11:

(1) # of depots: 2 depots attached to stations SH and TQ.

(2) # of rolling stock units storage of depots: A total of 20 rolling stock units are allocated for all instances.

(3) Passenger demand scenario: The passenger demand data along only a morning peak-hour passenger demand data from 6:45–10:15 (Scenario P2); only an off-peak-hour passenger demand data from 11:45–15:15 (Scenario P3).

(4) Computing time reduction = $\frac{SC-PC}{SC} \times 100\%$.

(5) The solution algorithm process is terminated when the computing time is over 7200 s in parallel computation and 28800 s in serial computation.

In particular, we consider the experiment instances on different planning time horizons in these tests. Specifically, the planning time horizon for instances 1–5 is assumed to be time interval [6:00, 23:00], while the planning time horizons for instances 6–7 are set as time intervals [6:45, 10:15] and [11:45, 15:15], respectively, in order to investigate the performances under the peak and off-peak hours precisely. For comparison in convenience, the computational results with instance features and computational performances are all displayed in Table 10 for large-scale instances 1–5, and in Table 11 for medium-scale instances 6–7, respectively. Moreover, since it is difficult to describe the change of cumulative passenger demand at each timestamp in the large-scale cases, a new indicator namely “secondary waiting passenger” is introduced to evaluate the passenger service quality, i.e., the column “secondary waiting passenger” in Tables 10–11 sums up the total number of passengers who cannot get on the first coming train.

Next, we aim to detail the performance analyses based on the results, and conclude some management insights with respect to different numbers of depots, rolling stock capacities and passenger demand distribution scenarios.

(1) Solution performance with respect to the number of depots. We observe from the computational results that, the operation flexibility of rolling stock can be improved to a great extent if more depots are built along the metro line. For instance, compared with instance 2 in which only one depot is involved, it is evident that the solutions with less secondary waiting passengers and objective values can be found when multiple depots are put into operations in instances 1 and 3. Especially, some extra rolling stock paths in instance 3 are also scheduled between SH and GZ stations to further satisfy the over-saturated passenger demand on this segment. A managerial insight thus might be that, multiple depots could be designed in the stage of constructing a metro line that possibly has complex passenger characteristics, which might further improve the passenger service under the flexible train composition modes.

(2) Solution performance with respect to the rolling stock capacities. To our knowledge, rolling stock with much larger capacities could reduce not only the total objective value but also the number of waiting passengers on platforms, which potentially simplifies the difficulty of metro operations. For instance, we consider two types of rolling stock with different capacities in instances 1 and 5 (i.e., A-type and B-type), where we also assume that the unit fixed costs for these two types are the same. It follows from the computational results that, compared with instance 1 that uses B-type rolling stock, there is a significant decrease for the number of secondary waiting passengers and the objective value that uses A-type rolling stock in instance 5. Thus, if the infrastructure conditions permit, it is much better to dispatch rolling stock with larger capacities so as to improve the passenger service level and reduce operation costs.

(3) Solution performance with different passenger demand scenarios. Similar to the performance in small-scale cases, both the generated train schedule and train composition modes could be dominated by the specific passenger demand scenarios. Let us consider instances 6 and 7 with passenger demand scenarios P2 and P3, respectively, in which the same condition regarding rolling stock is involved. In the results with Table 11, we observe that the generated trip services and train composition modes are different between the solutions of instance 6 and instance 7. More travelling trips in instance 6 are assigned with large train compositions in order to satisfy the over-saturated cumulative passenger demand (see Fig. 20(a)). While in instance 7 shown in Fig. 20(b), only 4 trips are scheduled with large compositions, aiming to process deadheading trips instead of carrying over-saturated passengers. According to these analyses, we draw that a good-performance operation scheme for peak-hour passenger demand is to apply large train compositions with proper headway adjustments when residual rolling stock is available. On the contrary, some trips can also be assigned with large compositions as additional deadheading trips if necessary according to the spatial-temporal distribution of the available rolling stock.

7. Conclusions and future research

In this study, aiming to obtain a passenger-oriented rolling stock scheduling scheme with the flexible composition mode, we formulated the problem of interest into an arc-based and an path-based network flow models, in which both passenger assignments and rolling stock circulations are represented by flow balance constraints with continuous passenger flow variables and binary space-time arc (path) selection variables. Based on the column generation method, a pricing sub-problem was designed based on the path-based restricted master problem, in order to reduce the difficulty of solution algorithms. Then a branch-and-price approach was developed to solve the proposed model. Finally, the performance of the models and algorithms were tested with a series of

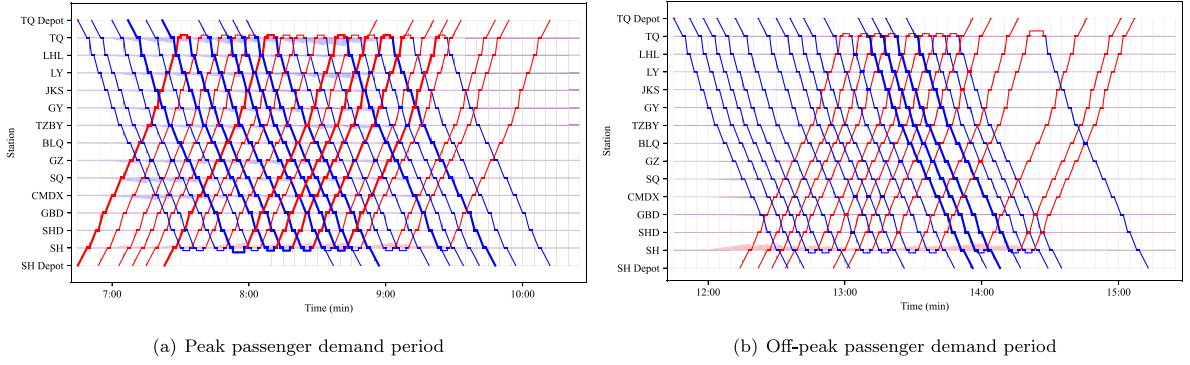


Fig. 20. Visualization of optimized solution during different passenger demand periods.

small-scale cases and large-scale real-world cases. The computational results demonstrate that the proposed methods can obtain high-quality solutions with less computing time than the Gurobi solver in both small-scale and large-scale instances. Simultaneously, the solution results can well balance train operation costs and passenger service qualities, which might help metro operators to make high-quality rolling stock scheduling plans.

From the perspective of operation and management insights, the train operation efficiency and passenger service level can be improved by concerning following aspects. (1) Compared with the fixed train composition mode, the flexible composition mode with multiple coupled rolling stock units can significantly reduce the total cost of rolling stock operations and passenger waiting time. Then, when multiple depots of rolling stock are built on the metro line, the effectiveness of the flexible composition mode is notable by using reasonable optimization method, as the capacity waste and total operation costs can be minimized to a great extent. (2) As for configurations of infrastructures, multiple turn-around tracks and rolling stock with large capacities along metro lines are encouraged to allocate if the operation condition permits, for improving the operation efficiency further. (3) As for the passenger demand, both trip frequencies and train compositions can be adjusted flexibly based on the passenger intensities with the proposed methods. At the same time, more large train compositions and trip frequencies can be scheduled to service stations and time periods with large cumulative passengers.

Further research will focus on the following aspects due to the limitations of our approaches. (1) This research still focuses on the problem of a metro line. A saving of operation and passenger waiting costs can be involved by scheduling rolling stock in different connecting lines to further improve the applicability of the proposed network-flow-based method. Thus, it is suggested to further consider such a rolling stock scheduling problem in larger instances with multiple lines and connecting stations. (2) More practical metro operation details can be integrated into the problem, such as crew scheduling or energy-efficient considerations. (3) A more efficient solution algorithm can be designed with more effective and tighter lower bounds to enhance the solution quality and efficiency.

CRediT authorship contribution statement

Entai Wang: Software, Visualization, Writing – original draft. **Lixing Yang:** Conceptualization, Methodology, Project administration, Supervision, Writing – review & editing. **Jiateng Yin:** Conceptualization, Investigation, Writing – review & editing. **Jinlei Zhang:** Supervision. **Ziyou Gao:** Funding acquisition, Project administration, Resources.

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Appendix. Experiments for different time discretization granularities

To our knowledge, different time discretization granularity could result in rounding errors of traveling and dwell time, which may affect the solution structures and objective values. However, too small and refined discretization granularity could result in larger space–time network with exponential growth in computing time. In this section, in order to test the balance between computing performance errors and time in our model, different time discretization granularities near the used value 60 s (i.e., 45 s, 60 s, 90 s, 120s) are employed additionally to investigate error performances with corresponding rounded traveling and dwell time, based on

Table 12
Computational performances in instances with different discretization granularities.

Discretization granularity	45 s	60 s	90 s	120 s
Solution feasibility	Yes	Yes	Yes	No
Objective value	4450.23	4865.58	5010.30	/
Rolling stock cost	3307.50	3570.00	3990.00	/
Passenger cost	1142.73	1295.58	1020.30	/
Computing time (s)	41.60	9.75	5.40	/
Objective Value Deviation (%)	0.00	+8.50	+11.10	/
Computing time Deviation (%)	0.00	-76.60	-87.00	/

small-scale instance ST_C2_P1. The comparison of the computational results by different discretization granularities can be referred to Table 12.

From the perspective of solution feasibility, when the discretization granularity is enlarged to 120 s, no feasible solution is obtained based on the rounding running/dwell time method. Then, we can observe that objective values go up with the growth of time granularities, and computing time go down with the growth of time granularities. Particularly, we can find that the benefits of computing time-saving (i.e., computing time deviation > 70%) get much higher than the benefits of objective values (i.e., objective value deviation < 10%) between the 45 s and 60 s granularities. However, when the time granularity is further increased to the 90 s, the benefit in computing time only decreases by about 10% with about a 3% increase in the objective function. In view of significant differences in computing time and relative small errors in objective values, the discretization granularity with 60 s is selected in this case for balancing the computing time and the solution accuracy.

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